Scene-Driven Multitask Parallel Attention Network for Building Extraction in High-Resolution Remote Sensing Images

Haonan Guo, Qian Shi, *Senior Member, IEEE*, Bo Du, *Senior Member, IEEE*, Liangpei Zhang, *Fellow, IEEE*, Dongzhi Wang, and Huaxiang Ding

***Abstract*— The application of convolutional neural networks has been shown to significantly improve the accuracy of building extraction from very high-resolution (VHR) remote sensing images. However, there exist so-called semantic gaps among different kinds of buildings due to the large intraclass variance of buildings, and most of the present-day methods are ineffective in extracting various buildings in large areas that cover different scenes, for example, urban villages and high-rise buildings, because existing building extraction strategies are the same for various scenes. With the improvement of the resolution of remote sensing images, it is feasible to improve the image interpretation based on the scene prior. However, this idea has not been fully utilized in building extraction from VHR remote sensing imagery. This study proposes a scene-driven multitask parallel attention convolutional network (MTPA-Net) to resolve these limitations. The proposed approach classifies the input image into multilabel scenes and further separately maps the buildings in pixel level under different scenes. In addition, a simple postprocessing method is applied to integrate the building extraction results and scene prior. Our proposed method does not require multimodel training and the network can learn in an end-to-end manner. The performance of our proposed method is evaluated on a data set that includes various urban and rural scenes with diverse landscapes. The experimental results show that the proposed MTPA-Net outperforms state-of-the-art algorithms by reducing misclassification areas and maintaining improved robustness.**

***Index Terms*— Building footprint extraction, deep learning, remote sensing image, scene driven.**

I. Introduction

AS THE material carriers of human production and life, buildings are essential indicators of urban expansion and population aggregation [1]. The building footprint extracted from remote sensing images constitutes a key factor in numerous geospatial applications such as urban planning [2], building model reconstruction [3], and mapping [4]. In particular, information such as the location, size, and number of buildings is indispensable for building-oriented applications including building change detection [5], real-time navigation, and 3-D urban modeling [6]. Such applications require high accuracy, regular updates, and different levels of automation. Traditionally, extracting building footprints was performed by manual interpretation and vectorization, which is arduous and time consuming. With the rapid development of sensors and the gradual improvement of remote sensing image resolution, very high-resolution (VHR) imagery provides opportunities to realize accurate and automatic extraction of buildings. However, building footprint extraction from VHR remote sensing images is still a challenging task, and the diversity of geometric shapes among buildings in combination with high complexity of scene forces researchers to consider specific solutions for each problem [7]. Therefore, developing effective and robust algorithms for building extraction has become a challenging and popular research topic in the field of remote sensing.

Over the last few decades, tremendous efforts have been made to develop various building extraction methods. Based on the data source, the building extraction methods mainly include optical image-based, nonoptical data-based, and data fusion-based methods. Optical remote sensing images, especially VHR remote sensing images, provide valuable information such as spectral, textural, and geometric information [8]. Traditionally, the algorithms that extract buildings from optical remote sensing images rely on features that can best represent a building such as color [9], spectrum [10], texture [11], shape [12], and context [13]. Among these features, the discrepancy of spectral features is directly reflected in the differences in the pixel values of different bands or adjacent pixels. However, spectral features are difficult to interpret because of the low discrepancy among buildings in VHR remote sensing images. Moreover, because these features can vary under different circumstances of light, sensor type, atmospheric condition, and roofing materials, leveraging spectral features alone has some limitations in discrimination among spectrally similar classes due to the homogeneity of spectral information in VHR imagery [4]. In addition to spectral features, geometric information such as texture [11], [14], shape [12], [15], and context [13] is capable of complementing spectral features

Manuscript received February 2, 2020; revised May 29, 2020 and July 10, 2020; accepted July 31, 2020. Date of publication August 19, 2020; date of current version April 22, 2021. This work was supported in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2019A1515011057; in part by the National Natural Science Foundation of China under Grant 61976234; in part by the Guangzhou Basic and Applied Basic Research Project; and in part by the Open Research Fund of National Key Laboratory of Surveying, Mapping and Remote Sensing Information Engineering, Wuhan University. *(Corresponding author: Qian Shi.)*

Haonan Guo and Qian Shi are with the School of Geography and Planning, Sun Yet-Sen University, Guangzhou 510275, China (e-mail: guohnwhu@163.com; shixi5@mail.sysu.edu.cn).

Bo Du is with the School of Computer Science, Wuhan University, Wuhan 430079, China (e-mail: gunspace@163.com).

Liangpei Zhang is with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China (e-mail: zlp62@whu.edu.cn).

Dongzhi Wang and Huaxiang Ding are with the Surveying and Mapping Institute Lands and Resource Department of Guangdong Province, Department of Natural Resources of Guangdong Province, Guangzhou 510500, China (e-mail: harry84@163.com; wydinghx@163.com).

Color versions of one or more of the figures in this article are available online at https://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2020.3014312

0196-2892 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.



Fig. 1. Examples of buildings with different structure and spatial distribution. (a) Urban village. (b) High-rise buildings. (c) Industrial buildings (d) Rural villages. (e) Urban village, industrial buildings and high-rise buildings.

by detecting abrupt gradient changes among adjacent pixels. However, owing to the variety of light and texture conditions of an image, generating a stable and generalized result in complex scenes where buildings are considerably diverse is still a challenging task [16]. Recently, the digital terrain model (DSM) generated by LiDAR, a rapidly developing technology, has shown its effectiveness in extracting ground objects [17]. The 3-D information provided by LiDAR is useful for realizing accurate building extraction. Additionally, the complementary information provided by multisource data, such as combining LiDAR data and optical images, is more discriminative in distinguishing buildings from nonbuilding objects [8] than single source data, although multisource data acquisition is usually costly.

Recently, the increasingly sophisticated state-of-the-art (SOTA) deep-learning methods provide alternative methods to study on automatic building footprint detection and building extraction. In addition to using handcrafted features, recent work has focused on applying convolution neural networks (CNNs) in the field of automatic building extraction [18]. CNN is a deep learning method that is capable of extracting hierarchical representations from raw image data and classifying images within a single model [19]. Because the features are learned from the data itself, CNN can avoid the subjectivity of handcrafted feature selection and is capable of providing improved performance. Although improvement can be achieved by designing deeper and more advanced networks [7], [20], [21], guaranteeing a reliable performance on a real-world data set that contains different scenes remains a challenge. The difficulty lies in the fact that the deepening of the network usually leads to an increase in parameters and computation, resulting in low training speed and massive memory usage. Additionally, sufficient training samples are needed to avoid overfitting in a complex model. Thus, a well-designed lightweight network may have better applicability than deep models in practice.

However, buildings can vary under different scales, structures, and distributions. As shown in Fig. 1(a), buildings in urban villages are low in height but high in density. In contrast, Fig. 1(b) shows that high-rise buildings are usually low in density but different from those of low-rise buildings. This indicates that there exist semantic gaps among different types of buildings due to the large intraclass variance of buildings in remote sensing image. Given a real-world data set that covers various types of buildings, designing a robust model that overcomes the semantic gaps and realizes accurate building extraction remains a challenging task that has not yet been solved. Taking this into account, some building extraction studies focus on extracting buildings under certain scenes (e.g., urban villages [10], [22], [23] and rural areas [9], [21]). However, these methods simply treat each building equally regardless of the variety of buildings in different scenes; the differences among buildings in different scenes have not been studied yet, as far as we know. Although buildings vary under different scales, structures, and distributions, buildings in similar scenes share similar characteristics. For example, in industrial areas [Fig. 1(c)], buildings are mainly composed of large-scale factories, which are usually low-rise buildings with large footprints. Moreover, the rooftops of industrial buildings are usually blue, green, and white, which are easy to recognize. Rural villages [Fig. 1(d)] are distributed on both sides of the road and are surrounded by fields and ponds. In addition to images that contain a single scene, there are also images that contain complex scenes in real-world data set. For example, as shown in Fig. 1(e), the image contains three types of buildings, which make up a complex scene consisting of urban villages, industrial buildings, and high-rise buildings. Based on the aforementioned examples, it can be inferred that the variances in structural characteristics and spatial distribution of buildings are mainly driven by their functions, which make it possible to recognize scene categories of an imagery based on building characteristics. Over the years, a lot of researches have been taken to recognize and analyze the scene category of VHR imagery, due to its wide application in land use mapping. Various methods such as probabilistic topic model [24], [25], feature coding [26], [27], and CNN-based [28], [29] models have been developed to categorize the scenes from VHR imagery. With the development of scene classification algorithms, it is feasible to realize "scene-object" collaborative image interpretation [30]. However, the variety of buildings among different scenes has not been fully studied yet. Although extensive studies have focused on extracting buildings in specific scenes, the effect of scenes on buildings has been largely ignored, and thus, the accuracy of these models in large-scale images with complex scenes was influenced to some extent.

In view of the conditions mentioned above, we introduce a novel scene-driven multitask building extraction network for VHR remote sensing images. This network is capable of overcoming the semantic gaps among different kinds of buildings by introducing the concept of "scene" and can realize end-to-end training and prediction. Our proposed network is composed of an encoder network, a classification network, and several decoder networks. As shown in Fig. 2, first, features in multiple levels of representations are extracted from an encoder network. Subsequently, a classification network classifies the extracted features into several different scene categories (e.g., urban villages, industrial buildings, low/mid-rise buildings, high-rise buildings, rural villages, low-density buildings, and high-density buildings), while several scene-specific decoder networks predict the prediction maps based on scene-prior information. Finally, a simple yet effective postprocessing method is performed to integrate the outputs of building extraction and scene classification results. It should be noted that the number of decoder networks is equal to the number of scene categories, and each decoder is trained only by images that contain the specific scene.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

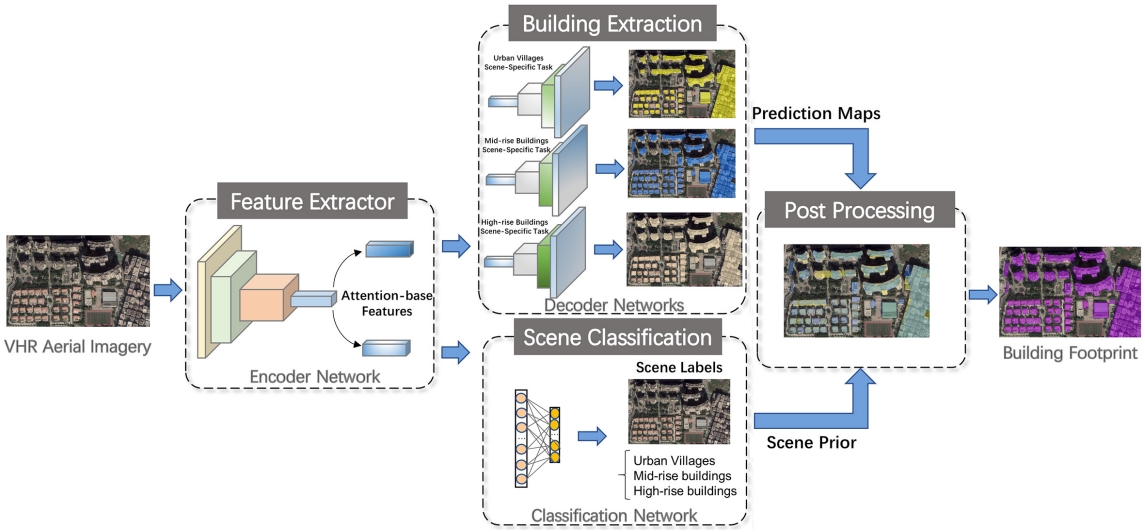


Fig. 2. Flowchart of the proposed method.

In this way, the proposed model can overcome the semantic gaps among different types of buildings by learning task-specific features in each scene-specific decoder network, while sharing expressive features extracted from the encoder network. In summary, the main contributions of this study are as follows.

1) We propose a new arrangement of a deep learning network named multitask parallel attention network (MTPA-Net) for building extraction by considering the scene prior, which aims to overcome the semantic gaps among buildings in different scenes. We also present a variant of our basic architecture, a single MTPA-Net, to verify the effectiveness of scene prior information in improving the accuracy of building extraction. The extensive experimental results on a real-world data set show that the proposed method outperforms seven SOTA building extraction methods by maintaining better robustness to buildings in various scenes.
2) We design a deep learning-based multitask architecture equipped with atrous convolution and attention mechanisms that enable the model to learn both task-shared and task-specific features in an end-to-end manner. The proposed architecture combines scene classification and scene-specific building extraction in a single model.
3) We introduce several lightweight SOTA soft-attention mechanisms to our proposed network. The attention mechanisms automatically determine "what" and "where" to focus on the shared features for the respective tasks. The results of the ablation experiments confirm that the proposed attention mechanisms improve the accuracy of building extraction and can be migrated to other building extraction networks.

The remainder of this article is organized as follows. Related work is described in Section II. In Section III, the details of the proposed MTPA-Net for VHR remote sensing image building extraction are introduced. The data set along with the experimental setup, results, and analysis are presented in Section IV. Finally, conclusions are presented in Section V.

II. Related Works

In recent decades, the extraction of buildings from remote sensing images has been widely studied owing to its broad applications. Based on the data source used for building extraction, the existing approaches can be categorized into three main categories: optical image-based, nonoptical data-based, and data fusion-based methods.

Traditional building extraction studies for optical remote sensing images usually exploit handcrafted features, such as color, spectrum, shape, texture, and shadow. For example, Huang and Zhang [13] proposed a morphological building index (MBI) for automatic building extraction from high-resolution imagery, which utilized morphological operators to present the spectral and spatial information of buildings such as brightness and contrast. They also proposed another object-based building extraction framework [31], in which spectral information and shadow information were exploited. In their study, a morphological shadow index was proposed to detect shadows, which constrained the outlines of the extracted building. However, using only spectral information has some limitations to fully express the building characteristics [4]. Taking this into account, some building extraction studies considered leveraging geometric information from remote sensing images. To solve the problem that buildings are more distinguishable in the geometric than in the spectral domain, Sirmacek and Unsalan [32] presented a building detection method based on scale-invariant feature transform and graph theory. Xia *et al.* [33] proposed a geometric building index in which geometric saliency was computed based on meaningful junctions that could describe the geometrical information of imagery. In addition to studies that use optical images as data sources, there are also studies that focus on extracting buildings from nonoptical data, such as LiDAR data. Recently, rapid

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

developing airborne LiDAR technology has opened up new ways to realize accurate building extraction. LiDAR point clouds provide 3-D information of the land surface, which is useful for extracting buildings [34]–[36]. Du *et al.* [34] combined grid-based and point-based features extracted from LiDAR data to realize robust building extraction. Zhang *et al.* [36] proposed a framework that continuously applies a progressive morphological filter, a region-growing algorithm, and a postprocessing method to extract buildings from LiDAR data. However, leveraging only a single data source has some limitations in fully expressing the building characteristics. Recently, an increasing number of studies have focused on using multisource data (e.g., LiDAR data and aerial images) to extract buildings [8], [37]. For example, Meng *et al.* [38] proposed an object-oriented classification method to detect residential buildings from LiDAR data and aerial images. Although fusing multisource data can improve the accuracy of building extraction, LiDAR data are more expensive than optical remote sensing images, thereby leading to the problem of data accessibility.

In addition to utilizing handcrafted features, deep feature-based methods have also attracted much attention in building extraction in recent years. Owing to the rapid development of deep CNNs, it has been demonstrated that deep feature-based methods can achieve impressive performance on processing remotely sensed imagery, such as image classification [28], [39] and object detection [40], [41]. Compared with the handcrafted feature-based method, CNNs are capable of extracting hierarchical semantically related representations from raw images without any prior knowledge. Moreover, CNNs can extract not only low/mid-level features but also high-level contextual features from remote sensing images. The task of semantic segmentation, which means making intensive predictions for each pixel, has attracted broad interest in the field of computer vision since Long *et al.* [42] proposed a fully convolutional network (FCN) in 2014. The FCN architecture replaced fully connected layers with convolutional layers. Moreover, the application of transposed convolutional layers and skip connections in an FCN enables feature maps to achieve the original resolution and make pixel-wise predictions. Based on the FCN diagram, several variants of FCN have been proposed to exploit multilevel information for more accurate predictions. A significant contribution comes from the encoder–decoder structure introduced in the U-Net [43] and SegNet [44] architecture. The encoder–decoder structure enables the network to maintain high-level semantic information while maintaining low-level detail information by transmitting multilevel semantic knowledge from the encoder to the decoder with multiple shortcuts. Additional improvements have been made by fusing multiscale features and enlarging the receptive field. Chen *et al.* [45] highlighted the importance of atrous convolutions in their proposed Deeplab network. With the design of atrous convolution, the receptive field of the network was improved without increasing the number of parameters. Moreover, their model also adopted a conditioned random field as a postprocessing method to refine their final prediction. Further improvement of the Deeplab network lies in designing an atrous spatial pyramid pooling (ASPP) module that includes multiple atrous convolutional layers [45], optimizing the ASPP structure [46], and introducing an encoder–decoder structure to a Deeplab network [47]. In addition, there are other variants of FCN, such as PSP-Net proposed by Zhao *et al*. [48], which is capable of aggregating multiscale context information to improve the ability to obtain global information.

The aforementioned FCNs have achieved outstanding performance not only in natural images but also in remote sensing imagery. Compared with extracting from natural images, extracting objects from remote sensing images is more challenging for two main reasons. First, ground objects in remote sensing images have larger intraclass variance, which makes it difficult to distinguish objects from the ground surface. Second, the location, size, and shape of ground surface objects greatly vary in different areas. Extensive research has been carried out to improve these SOTA models, making them suitable for building extraction. For example, Alshehhi *et al.* [49] designed a single patch-based CNN architecture to extract buildings and roads from remote sensing imagery. Based on the deep residual network, Xu *et al.* [50] proposed the Res-U-Net module based on the U-Net architecture. Ji *et al.* [51] developed the U-Net architecture and introduced a Siamese network, SiU-Net, in which the original imagery and its counterpart shared the same weights. They also created a high-quality multisource data set for building detection. Wei *et al.* [52] proposed an MA-FCN model to delineate the building footprint by voting from multiple models, and two postprocessing strategies were designed to refine the segmentation map. In CNN-based building extraction methods, it is challenging to balance the tradeoff between acquiring abundant context and detailed boundary information [18]. While an accurate boundary can be refined by extracting rich low-level features, contextual information is usually hard to obtain because it largely depends on the reception fields of high-level features. To capture and aggregate multiscale context information, Liu *et al.* [4] introduced a spatial residual inception module with large convolution kernels in their FCN. Liu *et al.* [53] designed inception-style downsampling modules in their deep encoder network to capture large reception fields.

Although the aforementioned methods may work to some extent, their limitations are that they merely treated the buildings in their study area equally, regardless of the semantic gaps among buildings that belong to different scenes. This may lead to over dependence on a large data set, reducing the transferability of their methods. Moreover, although it is feasible to capture rich contextual information and increase the inception fields by using larger convolution kernels or designing deeper networks, it usually leads to an exponential increment of parameters in the network, which is not effective in practical applications. Unlike the abovementioned deep feature-based method that directly applies FCN architectures to building extraction, we designed a multitask variant of FCN that combines scene classification and building extraction in a single network. In this way, the hierarchical features are fully utilized by different tasks, and the accuracy of the building extraction task can be improved by the scene prior knowledge provided by the classification task.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

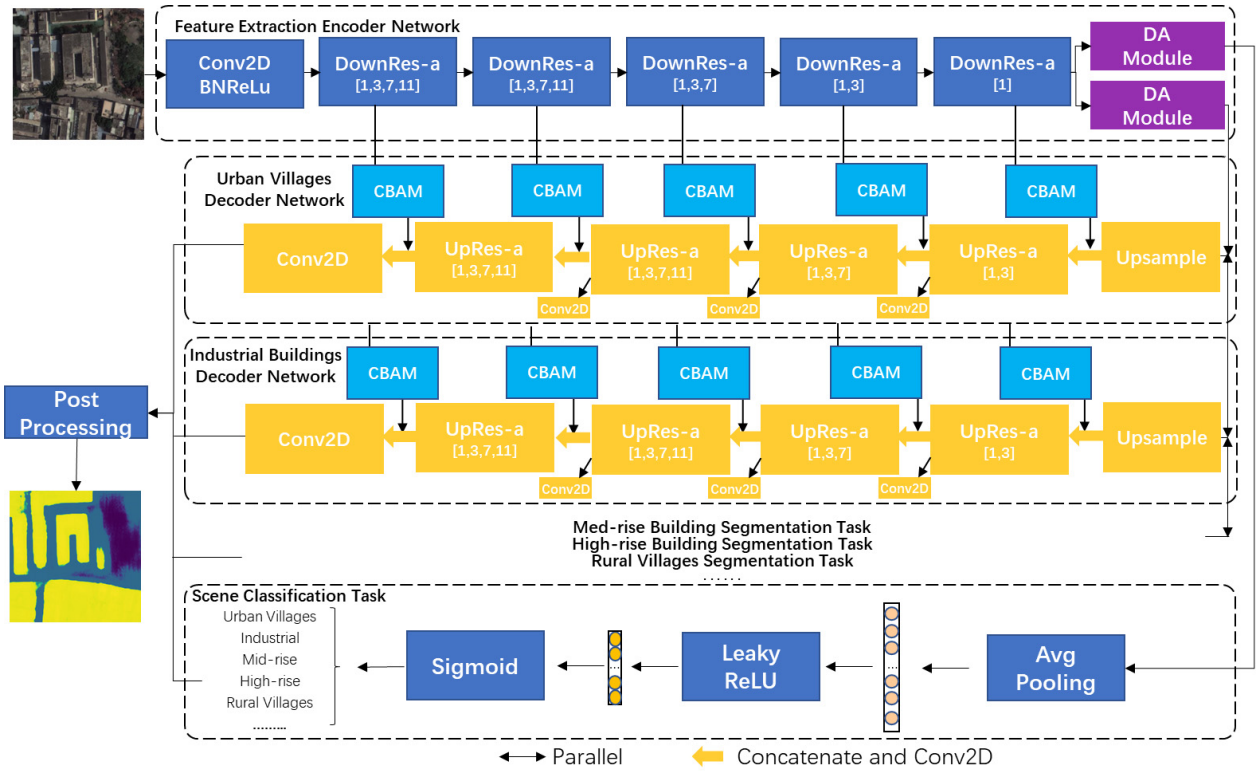


Fig. 3. Structure of the MTPA-Net.

III. Methodology

In semantic segmentation, the FCN and its variants have been successfully developed for building extraction. Inspired by these models, we propose an end-to-end variant of FCN called MTPA-Net in this study. As shown in Fig. 3, the proposed MTPA consists of three parts: an encoder network, a scene classification network, and scene-specific decoder networks. First, an encoder network is designed to guide the network to extract multiscale features with local and global contextual information. The extracted features are shared between the scene classification networks and decoder networks. The shared features are fed to two identical attention modules, the dual-attention module (in purple) and the convolutional block attention module (in light blue) to enhance the informative information and suppress irrelevant information. Subsequently, the low-resolution feature maps are classified into several different scene categories (e.g., urban villages, industrial buildings, low/mid-rise buildings, high-rise buildings, rural villages, low-density buildings, and high-density buildings) by a classification network. Meanwhile, each decoder network separately maps the multiscale features to full-input resolution feature maps for pixel-wise classification of buildings under different scenes. Finally, a simple postprocessing method is utilized to integrate the segmentation results and scene prior. In this section, we will introduce these three networks and the postprocessing method in succession; we will then present our single-task variant of the MTPA-Net.

A. Encoder Network

Extracting appropriate features from remote sensing images is one of the critical factors in determining the accuracy of building extraction. Our encoder network uses red, green, and blue bands from remote sensing images as the input data. The proposed encoder network, which consists of stacked layers of a residual downsampling module with atrous convolution (denoted as DownRes-a), is capable of extracting multiscale features from the input data. The architecture and parameters are as follows.

As shown in Fig. 3, the input of the encoder network first increases the number of feature layers by a convolution block that includes a 2-D convolution layer of kernel size $(1, 1)$, a batch normalization layer, and an activation layer. The kernel size of the convolution layer is chosen as 1 to avoid information loss from the input images by summarizing features across pixels with a larger kernel size. The activation layer is a nonlinear activation layer that endows a network with nonlinear ability. We chose Leaky ReLU as the activation function in our model instead of the widely used ReLU activation function, because neurons do not learn when ReLU function enters the negative range.

Then follow five DownRes-a modules. As is shown in Fig. 4, there are as many as four parallel branches of atrous convolutions in a DownRes-a module. Although a large reception field can be captured by designing larger convolution kernels, these kernels will lead to an exponential increment of parameters. Atrous convolution, which can increase the convolution kernel receptive field without losing any detailed information, is introduced to solve the proposed problem. Each branch corresponds to a different dilation rate determined by d_i. The output features of each branch have the same size and number of channels as the input. At the beginning of each branch, an atrous convolution is first adopted to extend the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

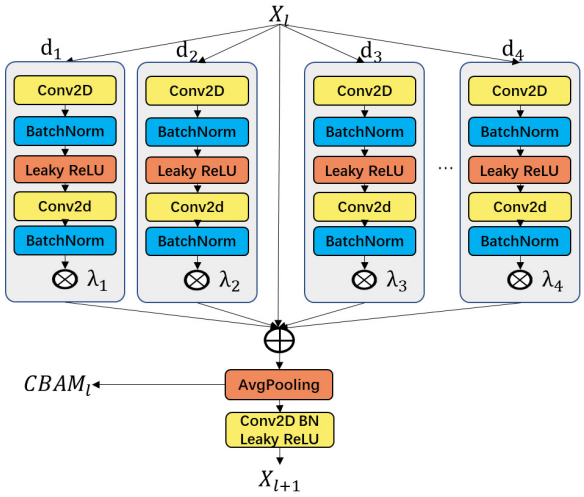


Fig. 4. Architecture of the DownRes-a module in the encoder network.

receptive field. The output channels of the atrous convolution layer are 1.5 times the amount of input channels. Then a convolution layer with a kernel size of (1, 1) and a dilation rate of 1 is adopted to integrate the features and reduce the number of channels to that of the input channel. Before summation of the outputs of each dilation branch and the initial input, we adopt trainable parameters λ_i to each branch to control the contribution of branches with different convolution rates to the output by a multiplication operation. It should be noted that we use a sigmoid function to limit the contribution rate between 0 and 1, and initialize λ_i to -2, which means that in the initial stage of training, the contribution rate of each branch is 0.12; the contribution rates are gradually optimized as the training continues. After that, the output of each dilation branch is added to the initial input to implement a residual structure, which helps to alleviate the problem of gradient disappearance and explosion. We adopt summation instead of concatenation because summation is easier to train. The DownRes-a module that integrates multiple reception information can be described as follows:

$$M_l = f_{l-1+} \sum_{i=1}^{N} \frac{1}{1+e^{-\lambda_i}} \times \cdot \mathrm{BN}\left(w^{c}_{1\times1,0}\left(\mathrm{Act}\left(\mathrm{BN}\left(w^{1.5\cdot h}_{3\times3,d_i}(f_{l-1})\right)\right)\right)\right) \quad (1)$$

where f_{l-1} denotes the input features and c denotes the number of bands of the input features. BN$(\cdot)$ and Act$(\cdot)$ indicate the batch normalization operation and Leaky ReLU activation function, respectively. $w^{1.5\cdot h}_{3\times3,d_i}$ denotes 3×3 convolutional kernels with d_i as the dilation rate and $1.5\cdot h$ as the number of bands of the output features. $w^{h}_{1\times1,0}$ indicates 1×1 convolutional kernels with no dilation, and the number of bands of the output features is equal to h.

The output of the summation layer is subjected to a pooling layer to halve the original size in spatial dimension, followed by a convolution block with a kernel size of (1, 1) to increase the feature channels to twice the amount of the input channels, which can be described as follows:

$$f_l = \mathrm{Act}\left(\mathrm{BN}\left(w^{2}_{1\times1,0}\cdot c(\mathrm{Pool}(M_l))\right)\right) \quad (2)$$

where Pool$(\cdot)$ and f_l denote the average pooling layer and the output of the DownRes-a module, respectively. $w^{2\cdot c}_{1\times1,0}$ denotes 1×1 convolutional kernels with no dilation, and the number of bands of the output features is equal to $2\cdot c$.

Feature maps extracted from the stacked DownRes-a modules have a dimension of 1/32 of the input size, which contains rich high-level features. However, there may still be some differences among features corresponding to the same label due to the large intraclass variance in remote sensing images. To obtain better feature representations, we introduce the dual-attention mechanism [54] (denoted as DA module) to explore accurate global contextual information by building associations among features. The structure of the DA module consists of two parts: a channel attention module and a positional attention module. The two attention modules capture long-range contextual information in the spatial and channel dimensions. Subsequently, a pixel-wise sum operation, a convolution layer, and a batch normalization layer are further applied to generate the final result.

1) Positional Attention Module: Given an input feature **I** with the shape of $F\times S\times U$, where F, S, and U denote the channel, height, and width of the input feature, respectively, we first feed the input feature to two separate convolution layers of kernel size (1, 1) to generate two new feature maps **A** and **B**. The two convolution layers are the same in structure but have separate parameters. The output channel of the two convolution layers is $(F/8)$. Then we reshape **A** to the shape of $N\times(F/8)$ and **B** to the shape of $(F/8)\times N$, where N is equal to $S\times U$. After that, we perform a matrix multiplication between **A** and **B** to generate a feature **M**. According to the principle of matrix multiplication, the shape of the feature **M** is $N\times N$. Subsequently, a softmax operation is applied to **M** and generates the positional attention map.

After acquiring the positional attention map, we feed the feature **I** to another convolution layer of kernel size (1, 1) and generate feature **P**. Features **P** and **I** have identical shapes. A matrix multiplication is then applied between the positional attention map and feature **P**, and the result is reshaped to the shape of $F\times S\times U$. Finally, we multiply it by a trainable scale parameter, which is initialized to zero, and perform an element-wise sum operation with feature **I** to obtain the final output.

2) Channel Attention Module: To further improve the feature representation and enhance the significance of interdependent features, we build a channel attention module to exploit the interdependence between channel maps. Apart from applying convolutional layers, the proposed channel attention module is similar to the positional attention module in the structure. We directly reshape the input feature **I** to the shape of $F\times N$ and apply matrix multiplication between **I** and the transpose of **I**. Then, a softmax layer is applied to generate the channel attention map. After that, we perform matrix multiplication between the transpose of the channel attention map and **I**, and then reshape the result to the shape of $F\times S\times U$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

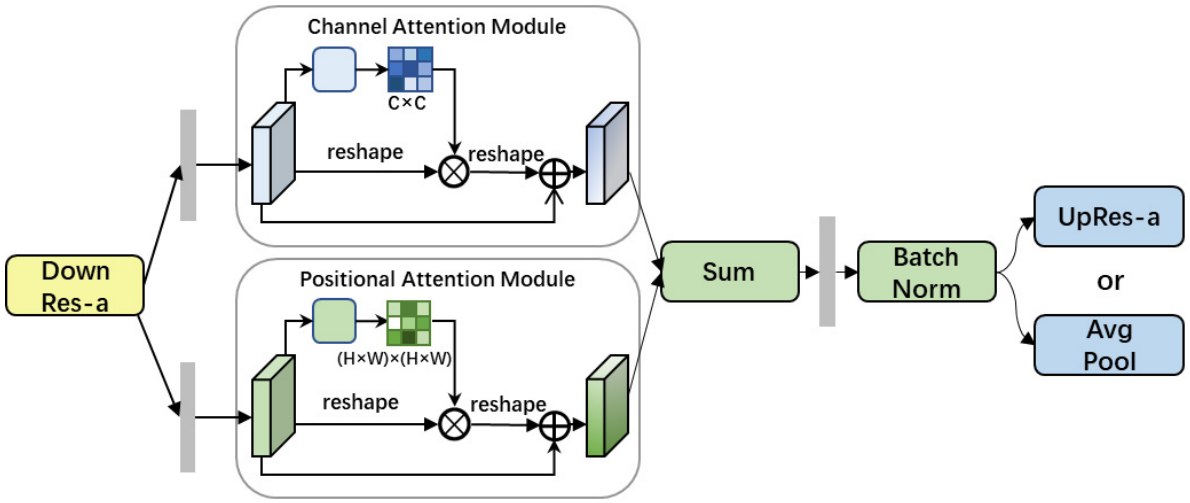


Fig. 5. Overview of the DA module.

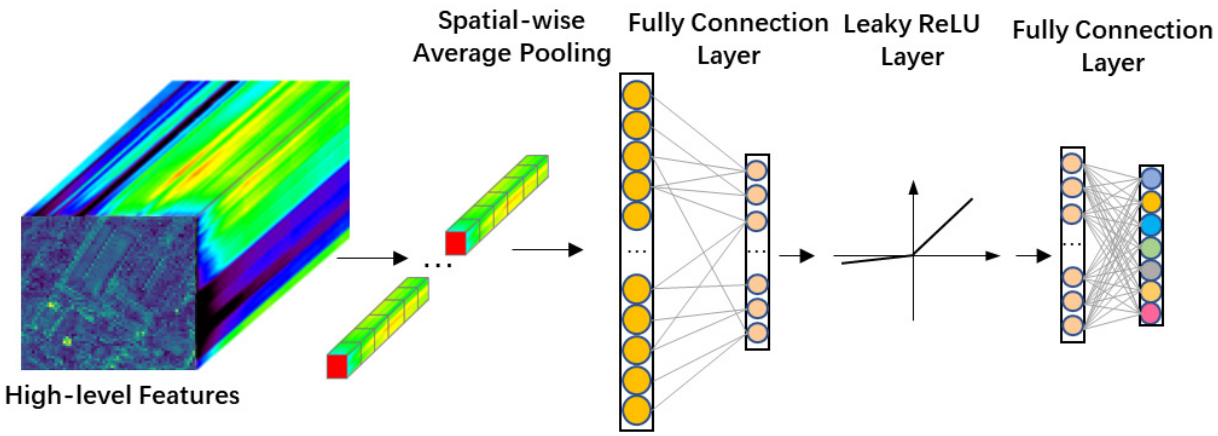


Fig. 6. Overview of the classification network.

Finally, we multiply it by a trainable scale parameter, which is initialized to zero, and perform an element-wise sum operation with **I** to obtain the final output.

B. Scene Classification Network

As is well known, buildings corresponding to different scene categories have different distributions and context relations, such as different surroundings between urban and rural buildings and building density of high-rise and low-rise buildings (see Fig. 1). If we separate building extraction strategies according to different types of scenes, it may result in an improved final extraction result. With this motivation, we first classify the high-level features extracted by the encoder network into several scenes based on the location, density, and height of buildings in the images. Specifically, the scene categories include urban villages, industrial buildings, low/mid-rise buildings, high-rise buildings, rural villages, low-density buildings, and high-density buildings. Among these scenes, urban villages and industrial, low/mid-rise, and high-rise buildings are buildings that are located in urban areas, where the percentage of impervious surface areas is higher than 70%. Low-density and high-density buildings are located in urban fringe areas, where the percentage of impervious surface areas is between 30% and 70%. Rural villages are buildings located in rural areas, where the percentage of impervious surface areas is lower than 30%. It should be noted that the proposed algorithm can be applied to scene categories different from those in our experiment. Considering that there may be more than one scene in a single image, we adopt the multilabel image classification method. The structure of the proposed scene classification is shown in Fig. 6. The input of our multilabel classification network is the high-level features extracted by the encoder network, which has 256 channels and is 1/32 of the initial input image size. First, we feed the features into an average pooling layer to compute the global average over the spatial dimensions and generate a 1-D vector of length 256. Then, we connect the vector to a fully connected layer with a weight parameter of 256×64. It should be noted that a dropout strategy was adopted to avoid overfitting. Finally, we adopt a Leaky ReLU activation function to improve the nonlinear learning ability of the classifier, followed by a fully connected layer. The output of the classifier is a weight vector, whose length corresponds to the number of classes.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

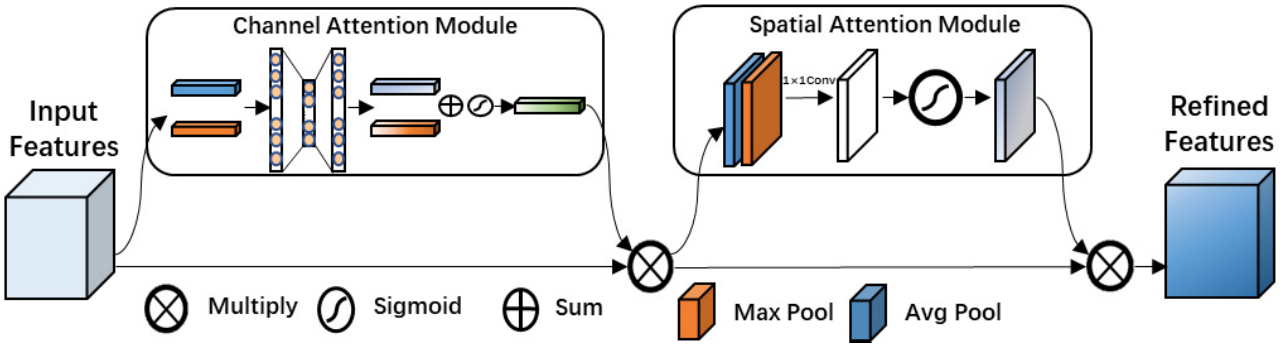


Fig. 7. Overview of the convolution block attention module.

The weight vector is fed to a sigmoid function and generates the final multilabel prediction probability.

C. Decoder Network

In the building extraction task, we feed the feature maps extracted by the encoder network to different decoder networks that correspond to different scenes. The decoder networks are capable of mapping the low-resolution extracted by the encoder network to full input resolution feature maps and makes pixel-wise classification. Each decoder network has the same structure but independent parameters. The number of the decoder network is equal to the number of scene categories. The architecture and parameters of a single decoder network are as follows.

In the decoder network, upsampling is implemented through a bilinear interpolation layer, which doubles the size of the input feature maps while maintaining the number of input channels. First, the high-level feature maps extracted from the encoder network are fed into a bilinear interpolation layer followed by a Combine layer. In the concatenation and convolution layer, denoted as the Combine layer, the features from the encoder and decoder parts are integrated and channels of features are compressed to a desired size by a convolution layer with a kernel size of (1, 1). It should be noted that before concatenation, the feature maps from the encoder network are fed to a convolution block attention module [55] (CBAM) with features from the decoder part to improve the representation in both the channel and spatial dimensions.

As shown in Fig. 7, we sequentially fed the input features to the channel attention modules and the spatial attention modules to enable the branches to learn "what" and "where" to focus. First, we produce a channel attention map through a channel attention module, which enables the network to capture "what" to see in the channel dimension.

In the channel attention module, the input features are mapped to two different vectors by a max-pooling layer and an average-pooling layer. Subsequently, both vectors are forwarded to a shared network to further produce channel attention maps. The shared network consists of a multilayer perception with one hidden layer. The size of the hidden layers is set to 1/16 of the channels of the input features to reduce the parameter burden. Lastly, we perform a summation operation to merge two channel attention maps, followed by a sigmoid function that generates the final attention map. An element-wise multiplication is performed between the input features and the 1-D channel attention map in the channel dimension.

Besides focusing on "what" to see, we further pass the results through a spatial attention module to generate the spatial attention maps, which facilitate focusing on "where" to see. To compute the spatial attention map, we first perform max-pooling and average-pooling operations along the channel axis to generate two feature maps. Then we concatenate the two feature maps and apply a convolution layer to produce spatial attention maps, followed by a sigmoid function that limits the attention values between 0 and 1. Lastly, we perform a multiplication operation between the features and the 2-D spatial attention map in the spatial dimension.

By transmitting the rich low/mid-level features into the decoder part through attention modules, the CBAM units emphasize the meaningful information while suppressing the meaningless features. Because the CBAM unit is lightweight, it can be widely embedded into our models for the transmission of low/mid-level features, which means that each of our decoder network owns respective CBAM units. It enables each decoder network to independently emphasize and compress the rich information extracted from the encoder according to each scene.

Then, the output of the first Combine layer is fed into stacked layers of improved residual blocks (UpRes-a) and Combine modules. As shown in Fig. 8, the UpRes-a block is similar to the DownRes-a block in structure, except that the pooling layer is replaced with an upsampling layer in UpRes-a after the summation of the outputs of each dilation branch and the input features. Moreover, unlike the DownRes-a block, the UpRes-a block does not have an intermediate output, and the number of channels is halved by the last convolution layer with a kernel size of (1, 1).

It should be stressed that apart from the main branch to train the final classifier in the decoder network, three more classifiers are applied after the first, second, and third UpRes-a blocks. These auxiliary classifiers can help to optimize the learning process in the backpropagation stage by passing through all previous layers, while the master branch loss takes the most responsibility. Weights are used to balance auxiliary loss.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

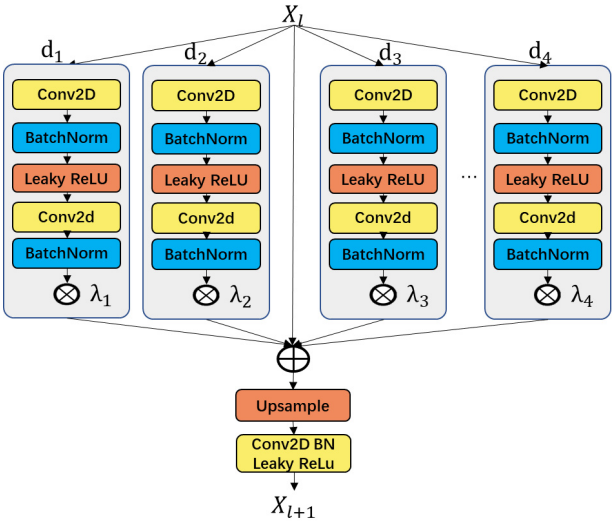


Fig. 8. Architecture of the UpRes-a block in the decoder network.

D. Postprocessing

Because each decoder network outputs a prediction map, a postprocessing process is proposed to integrate the results from the multilabel scene classification task and building extraction tasks. Given the prediction maps set $A = \{A_1, A_2, \ldots, A_N\}, A_n \in \mathbb{R}^{H\times W}$, and predicted scene categories $B = \{B_1, B_2, \ldots, B_N\}, B_n \in \{0, 1\}$, where N is equal to categories of scene classification, we first apply a sigmoid operation on the prediction maps set $\mathbf{A}$

$$C = \frac{1}{1 + e^{-A}} \tag{3}$$

where C represents the confidence maps set corresponding to different scenes, ranging from 0 to 1. The closer the confidence is to 1, the more likely the pixel belongs to the building category.

Subsequently, we perform element-wise multiplication in the channel axis between the binary scene categories and prediction maps, followed by an average operation to integrate the building extraction results and the scene classification result $S \in \mathbb{R}^{H\times W}$

$$\mathrm{S} = \frac{\sum_{i=1}^{N}(B_i \cdot C_i)}{\sum_{i=1}^{N} B_i}. \tag{4}$$

E. Single-Task MTPA-Net

We also designed a single-task variant of the MTPA-Net, named Single-MTPA-Net. In this version of the MTPA-Net, the scene classification network is removed and scene-specific decoder networks are replaced with a single decoder network. As shown in Table I, the encoder network (Layers 1–9) and the decoder network (Layers 10–30) comprise a simple yet effective encoder–decoder architecture based on FCN. Suppose that the shape of the input image is $3 \times H \times W$, where H and W denote the height and width of the image, respectively, the shape of each layer's output features is shown in Table I, column 3. Specifically, the encoder network extracts multiscale global and local contextual features by stacked layers of DownRes-a modules. Then, CBAM and DA module are able to capture long-range and short-range contextual information and learn "what" and "where" to focus on in the positional axis and channel axis, respectively. The architecture of the decoder network in a Single-MTPA-Net is the same as that of MTPA-Net's decoder network. However, in the encoder network, a DA module and a batch normalization layer that are connected to the scene classification task are removed as nuisance parameters.

TABLE I
DETAILS OF THE SINGLE-MTPA-NET

Layer#	Layer Type	Output Shape
Input		3×H×W
1	Conv2D(input,f=16,k=1,d=1,s=1)	16×H×W
2	BatchNorm(Layers1)	16×H×W
3	LeakyReLu(Layers2)	16×H×W
4	DownRes-a(Layers3,f=32,k=3,d={1,3,7,11},s=1)	32×H/2×W/2
5	DownRes-a(Layers4,f=64,k=3,d={1,3,7,11},s=1)	64×H/4×W/4
6	DownRes-a(Layers5,f=128,k=3,d={1,3,7},s=1)	128×H/8×W/8
7	DownRes-a(Layers6,f=256,k=3,d={1,3},s=1)	256×H/16×W/16
8	DownRes-a(Layers7,f=512,k=3,d={1,3,},s=1)	512×H/32×W/32
9	DAModule(Layers8,f=256)	256×H/32×W/32
10	Upsample(Layers9,f=256)	256×H/16×W/16
11	BatchNorm(Layers9)	256×H/16×W/16
12	CBAM(Layer8,f=256)	256×H/16×W/16
13	BatchNorm(Layer12)	256×H/16×W/16
14	Combine(Layer11&13,f=256)	256×H/16×W/16
15	UpRes-a(Layers14,f=128,k=3,d={1,3},s=1)	128×H/8×W/8
16	CBAM(Layer7,f=128)	128×H/8×W/8
17	BatchNorm(Layer16)	128×H/8×W/8
18	Combine(Layer15&17,f=128)	128×H/8×W/8
19	UpRes-a(Layers18,f=64,k=3,d={1,3,7},s=1)	64×H/4×W/4
20	CBAM(Layer6,f=64)	64×H/4×W/4
21	BatchNorm(Layer20)	64×H/4×W/4
22	Combine(Layer19&21,f=64)	64×H/4×W/4
23	UpRes-a(Layers22,f=32,k=3,d={1,3,7,11},s=1)	32×H/2×W/2
24	CBAM(Layer5,f=32)	32×H/2×W/2
25	BatchNorm(Layer24)	32×H/2×W/2
26	Combine(Layer23&25,f=32)	32×H/2×W/2
27	UpRes-a(Layers26,f=16,k=3,d={1,3,7,11},s=1)	16×H×W
28	CBAM(Layer4,f=16)	16×H×W
29	BatchNorm(Layer28)	16×H×W
30	Combine(Layer27&29,f=16)	16×H×W
31	Conv2D(Layer30,f=1,k=1,d=1,s=1)	1×H×W

IV. EXPERIMENTS AND RESULT ANALYSIS

A. Data Set Description

The performance of deep learning models relies on large-scale learning samples. To evaluate the performance of our proposed network, high-resolution remote sensing images with a coverage of 48.5 km^2 are used as experimental data. The data set includes Google Earth VHR images that cover urban areas in Shenzhen, China, and rural areas in Maoming, China. Moreover, we also implement our data set with images of urban fringe areas from the Inria Aerial Image Labeling Data set [56].

The urban study area is located in Shenzhen, Guangdong Province, China, which covers 7 km^2 with a resolution

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

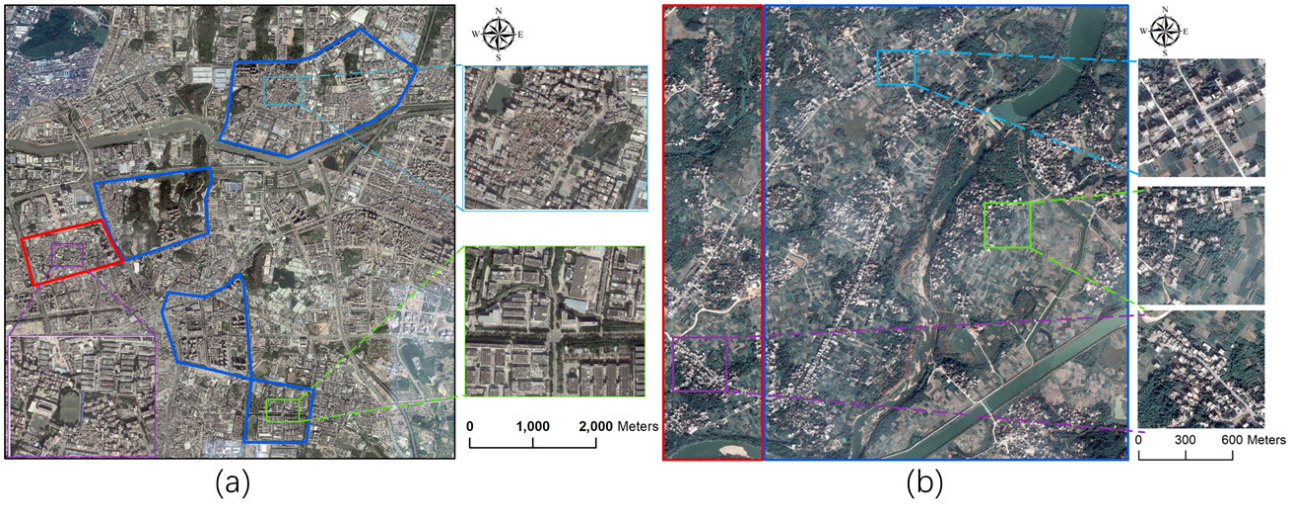


Fig. 9. Images that cover building areas in the aerial data set. (a) Urban study area located in Shenzhen. (b) Rural study area located in Dianbai. The areas in the blue box are used for training, and the area in the red box is used for testing.

TABLE II
DETAILS OF OUR DATA SET

District	Resolution	Area(km^2)	Source	Training	Test
Shenzhen	0.28m	7	Google Earth	1024	200
Dianbai	0.28m	10	Google Earth	600	100
Austin	0.3m	15.75	Inria Dataset	600	100
Chicago	0.3m	15.75	Inria Dataset	600	100
	Total	48.5		2824	500

of 0.28 m. As a pioneer of China's reform and opening-up policy, Shenzhen is the first city in China to be fully urbanized; thus, it can best represent urban areas. The percentage of impervious surface areas in urban areas is higher than 70%. The authorities are committed to building an intelligent identification system of urban elements by evaluating VHR remote sensing images, mapping the distribution of buildings, and differentiating various types of buildings. Although the evaluation of the distribution of buildings in this area is of great significance to the optimization of urban planning, it also presents a challenge in detecting large-scale buildings: building density in this area is rather high. Moreover, there are many different scenes in the area. Google Earth VHR images were captured between 2018 and 2019. As shown in Fig. 9, the urban study area is chosen in Bao'an District, Shenzhen, and divided into blocks through roads. Five blocks of images are selected randomly as our urban study area, which covers 7 km^2 and occupies large residual areas, contiguous industrial areas, and versatile mid/high-rise buildings. We use four of these images as training datasets and the rest as test datasets. We then manually edited the outline of the buildings in the study area and divided the building labels into four categories, including urban villages, industrial buildings, low/mid-rise buildings, and high-rise buildings. As can be seen in Fig. 10(a)–(d), urban villages have very dense residual areas that are low in height. Industrial buildings are buildings that are mainly for industrial use. Generally, industrial buildings are continuous and their roofs are painted with blue and white for heat dissipation. Excluding urban villages and industrial buildings, the buildings are classified into low/mid-rise buildings and high-rise buildings based on the buildings' height.

In the rural area, we chose an area from Dianbai, Maoming City, Guangdong Province, China, which contains various architectures and types of rural villages with different colors, architecture, and size. In rural areas, the percentage of impervious surface areas is lower than 30%. The imagery of Maoming covers 10 km^2 with a resolution of 0.28 m and is divided into training and test areas in the ratio of 4:1. After that, we manually edited the building outline in the rural study area. In addition, to further validate the effectiveness of our algorithm, we also supplemented our data set with an open-source building data set collected from the Inria data set. The Inria data set includes 180 tiles of aerial images from Austin, Chicago, Kispa, Vienna, and Tyrol. We chose parts of the Inria data set instead of fully utilizing it, to maintain the sample balance of the chosen urban and rural study area. We selected urban fringe areas in Austin and Chicago as the study area, where the percentage of impervious surface areas is between 30% and 70%. Six tiles of images and their

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

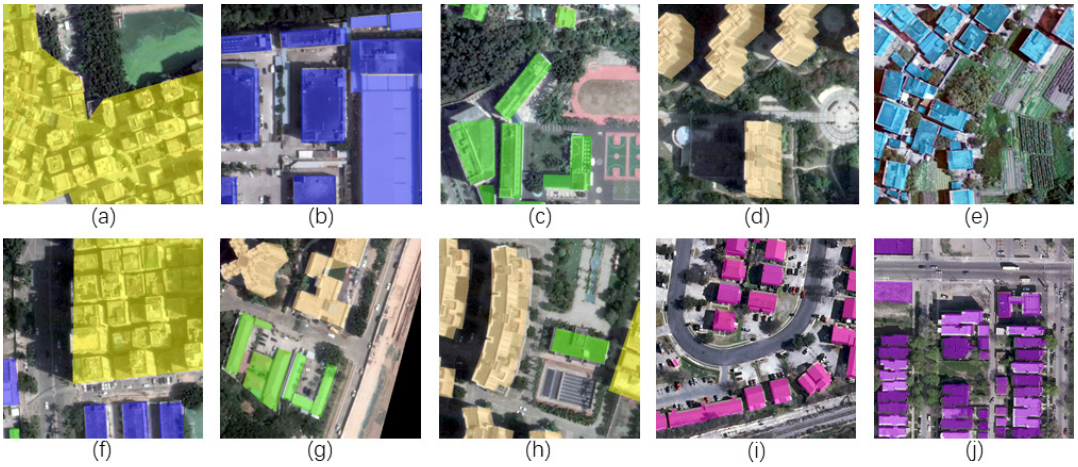


Fig. 10. Examples of the aerial imagery data set under different scenes with different architectures, purposes, and colors. (a) Urban Villages. (b) Industrial buildings. (c) Low/mid-rise buildings. (d) High-rise buildings (e) Rural villages. (f) Mixture of urban villages and industrial buildings. (g) Low/mid-rise buildings and High-rise buildings. (h) Urban villages, low/mid-rise buildings, and high-rise buildings. (i) Low-density buildings in urban fringe area. (j) High-density buildings in urban fringe area.

corresponding building footprints of Austin and Chicago were chosen, respectively, and were divided into two parts: six tiles for constructing the training set and one tile for the test set. Based on the density of buildings, we defined the buildings in Austin as low-density buildings and the buildings in Chicago as high-density buildings.

Finally, each image was cropped to create a sequence of 512×512 pixel patches with an overlap of 100 pixels. As a result, our data set includes 2824 tiles for training and 500 for testing the effectiveness of the algorithm, with an average resolution of 0.29 m. These patches are the processing units of our proposed method. During the training process, the network was trained only on the training set. After the model converged, the performance of the network was evaluated on the test set. Examples of our data set are shown in Fig. 10. It should be noted that contrary to the current openly available data set, labels in our data set include not only the building outline but also the scene category corresponding to buildings, which can realize scene-driven building extraction. In total, there were seven classes of buildings in our data set including urban villages, industrial buildings, low/mid-rise buildings, high-rise buildings, rural villages, low-density buildings, and high-density buildings. Among these scene categories, urban villages, industrial buildings, low/mid-rise buildings, and high-rise buildings were extracted from urban areas in Shenzhen. Rural villages were extracted from rural areas in Dianbai. Low-density and high-density buildings were extracted from urban fringe areas in Austin and Chicago. More details of our dataset are shown in Table II.

B. Experimental Details

Deep learning is a data-dependent method that is susceptible to overfitting. To avoid overfitting, we adopted data augmentation so that the batch of images was always different. Before data augmentation, we first matched the histogram of the image in each region to maintain the consistency of the image data distribution in each region. After that, each pair of image and label was rotated at a random angle between 0° and 30°. The angle was selected between 0° and 30° to balance the robustness of the model and image information preservation. Finally, we applied scaling operation according to a random factor between 0.9 and 1.1. If the scaling factor was smaller than 1, a zoom/in operation was applied to the image. Taking 0.9 as an example, a subimage of $460 \times 460 (512 \times 0.9)$ pixels was randomly clipped from the image of 512×512 pixels. Then the subimage was resized to 512×512 pixels using bilinear interpolation. If the scaling factor was larger than 1, a zoom-out operation was applied to the image. Taking 1.1 as an example, the image was resized to 460×460 pixels, and the left parts were filled in with zero paddings. We also adopted a random horizontal/vertical flip as an additional data augmentation routine. After augmentation, we normalized the images and standardized them with the precalculated mean and variance. At the inference stage, we predicted the prediction maps on test data set without applying data augmentation.

The proposed MTPA-Net was implemented using PyTorch [57] on two NVIDIA RTX 2080Ti GPUs. The network was trained with the SOTA Ranger optimizer, a synergistic optimizer combining Rectified Adam [58] and Look-ahead strategy [59], which yielded high scores for different categories on public leaderboards. The initial learning rate was set to 0.01, with a weight decay of 0.01. We adopted the learning rate warm-up strategy in the first five epochs of our training to improve the stability of model learning, which means that the learning rate of the first epoch was set to one-fifth of the initial learning rate and increased 0.002 per epoch until the learning rate reached 0.01. The total epoch number was set at 200, and the learning rate was reduced to one-tenth every 40 epochs. The weight parameters in the MTPA-Net were initialized with the method described in [60] using a normal distribution, and the bias parameters were initialized with zeros. For fully connected layers in the MTPA-Net, we initialized the weight data using a simple normal distribution with 0 as the mean value and 0.01 as the standard deviation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

TABLE III
PERFORMANCE EVALUATION OF SCENE CLASSIFICATION

Scene Category	Accuracy	Precision	Recall	F1-score
Urban Village	0.978	0.951	0.941	0.946
Industrial Building	0.934	0.843	0.778	0.809
Low/Mid-rise Building	0.97	0.824	0.757	0.789
High-rise Building	0.98	0.88	0.759	0.815
Rural Villages	0.994	1	0.971	0.985
Low-density Buildings	0.99	0.961	0.99	0.975
High-Density Buildings	0.996	1	0.98	0.99
Overall	0.977	0.914	0.943	0.928

In multitask learning networks, determining the correct balance between different classes is important for training multiple tasks. To test the efficiency of our method, we trained the MTPA-Net with pretrained parameters and the frozen encoder network. Specifically, in the training stage, we first trained the Single-MTPA-Net. R-G-B composite images were fed into a Single-MTPA-Net to generate the prediction maps. The Lovasz-Softmax loss [61], which has been proved to outperform the cross entropy loss in wide experiments, was used as the loss function to measure the gap between the ground truth and the prediction map, and to update the network parameters through backpropagation. After obtaining the trained Single-MTPA-Net, the parameters in the encoder network and the decoder networks of the MTPA-Net were initialized with the pretrained Single-MTPA-Net. Then, the network was trained with a frozen encoder network to avoid training imbalance caused by scene category imbalance. We adopted binary cross entropy loss for the scene classification task, which can be described as

$$L_c = -\frac{1}{K}\sum_{i=1}^{K} l_i \cdot \log(p_i) + (1 - l_i) \cdot \log(1 - p_i) \tag{5}$$

where K denotes the number of scene categories. p_i, which is derived from the sigmoid function, denotes the probability that the input image belongs to scene category i. l_i is the label of the input image that belongs to scene i.

The Lovasz-Softmax loss [61] was chosen for evaluating building prediction maps. We adopted weights to balance the loss between the main branches and auxiliary classifiers. The loss in building extraction tasks can be described as

$$L_s = \sum_{i=1}^{K} w_1 \cdot \mathrm{LS}(y_i, g_i) + w_2 \cdot \mathrm{LS}\left(y_i^{256}, g_i^{256}\right) + w_3 \cdot \mathrm{LS}\left(y_i^{128}, g_i^{128}\right) + w_4 \cdot \mathrm{LS}\left(y_i^{64}, g_i^{64}\right) \tag{6}$$

where LS$(\cdot)$ denotes the formula of the Lovasz-Softmax loss. y_i^j is the auxiliary classification map with the shape of $j \times j (j \in \{256, 128, 64\})$ from the tasks corresponding to the scene category i. g_i^j is the ground truth label with the shape of $j \times j (j \in \{256, 128, 64\})$, which is downsampled by bilinear interpolation. w denotes the weights designed to balance losses from the main branch and auxiliary classifiers. w_1, w_2, w_3, and w_4 are set to 0.7, 0.1, 0.1, and 0.1, respectively.

We assessed the performance of scene classification and building extraction on the test set using various evaluation metrics.

C. Quantitative Evaluation

In this study, five commonly used metrics, namely overall accuracy, precision, recall, F1 score, and intersection over union (IoU), were used to quantitatively evaluate the performance of the proposed MTAP-Net and comparative methods from different aspects.

The confusion matrix between the prediction maps and ground truth labels was calculated first, which determined the number of four classifying conditions on pixel-level: true building prediction on a positive sample (TP), false building prediction on a positive sample (FP), true building prediction on a negative sample (TN), and false building prediction on a negative sample (FN). Overall accuracy represents the percentage of correctly classified pixels (summation of TP and FN) over the total number of pixels. Precision is the ratio of TP in the entire positive prediction; recall indicates the proportion of TP over the entire positive samples. F1 score, the weighted average of precision and recall, takes both FP and FN into consideration, and the IoU score, an instance-level metric which is also known as the Jaccard similarity coefficient, represents the intersection of the ground truth and classification map over their union. The aforementioned metrics can be defined as

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{FN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FR} + \mathrm{FN}} \tag{7}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{8}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{9}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN} + \mathrm{FP}} \tag{10}$$

$$\mathrm{F1\ score} = 2 * \frac{\mathrm{Precision} * \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}. \tag{11}$$

D. Results and Analysis

1) Scene Classification: After obtaining the trained MTPA-Net, we assessed the performance of multilabel scene classification using overall accuracy, precision, recall, and F1 score in the test data set. The performance of scene classification in our proposed MTPA-Net is presented in Table III.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
CONFUSION MATRIX OF SCENE CLASSIFICATION FOR THE TEST DATA SET

Scene Category	Confusion Matrix	Positive Label	Negative Label	Precision(%)
Urban Village	Positive Prediction	96	5	95.05
	Negative Prediction	6	393	
Industrial Building	Positive Prediction	70	13	84.34
	Negative Prediction	20	397	
Low/Mid-rise Building	Positive Prediction	28	6	82.35
	Negative Prediction	9	457	
High-rise Building	Positive Prediction	22	3	88
	Negative Prediction	7	468	
Rural Villages	Positive Prediction	101	0	100
	Negative Prediction	3	396	
Low-density Buildings	Positive Prediction	98	4	96.08
	Negative Prediction	1	397	
High-density Buildings	Positive Prediction	97	0	100
	Negative Prediction	2	401	

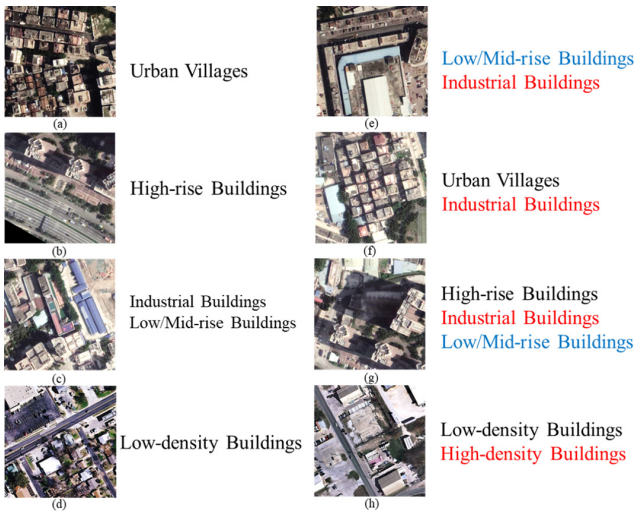


Fig. 11. Some of the scene classification results of the MTPA-Net. Labels in black denote true prediction on a positive label. Labels in blue denote false prediction on a negative label. Label in red denote false prediction on a positive label.

As can be seen in Table III, the overall accuracy, precision, recall, and F1 score in the test data set are 97.7%, 91.4%, 94.3%, and 92.8%, respectively, which confirm that the classification network of the MTPA-Net is an effective approach for VHR imagery multilabel scene classification. The confusion matrix of different scene categories is listed in Table IV. Considering that we adopted multilabel classification, which can also be seen as seven separate binary classifications, we applied seven 2×2 confusion matrices in Table IV. Overall, the classification results of most classes perform well. The F1 score for high-density buildings, 0.99, was the best among all the scene categories. Low-density buildings, urban villages, and rural villages also achieved good performances with only 4.9%, 10.3%, and 2.91% of the scenes are misclassified, respectively. This demonstrates that buildings with large differences in architecture, structure, and landscape are easy to distinguish. For example, as is shown in Fig. 11(a), urban villages usually have high density and are continuous in structure. There is, however, some confusion in highly mixed scenes, especially in industrial buildings and low/mid-rise was buildings. The precision and recall of low/mid-rise buildings are 82.4% and 75.7%, respectively, which indicate that low/mid-rise buildings are difficult to distinguish. This can be explained by low/mid-rise buildings corresponding to the buildings lower than eight floors except for urban villages and the industrial buildings; thus, the intraclass difference of mid-rise buildings is large compared with urban villages and high-rise buildings. Moreover, the scene belonging to industrial buildings is misclassified as negative prediction, which is because some images in the test data set only include a small proportion of industrial buildings, resulting in low confidence in prediction. To facilitate visual inspection, some of the classification results are shown in Fig. 11.

2) Building Extraction: The quantitative building extraction results of Single-MTPA-Net and MTPA-Net are shown in Table V. The bolded values in Table V represent the higher value of the evaluation metrics between Single-MTPA-Net and MTPA-Net. As can be seen in Table V, adding scene prior to the process of building extraction improves performance in extracting high-rise buildings, rural villages, low-density buildings, and high-density buildings. Compared with the Single-MTPA-Net, the MTPA-Net achieved approximately 1.5%, 5%, 3.8%, and 5.2% relative increase in the value of IoU on high-rise buildings, rural villages, low-density buildings, and high-density buildings, respectively. The results demonstrate the importance of incorporating scene prior to the process of building extraction. There was a slight decrease in the F1 score of industrial buildings (80.4%–80.25%) and mid-rise buildings (87.43%–86.37%). The reason for the performance can be explained as follows. First, the MTPA-Net is effective in extracting buildings from a single scene by reducing false predictions, as it achieved great improvement in rural villages, low-density buildings, and high-density buildings. Second, industrial buildings and mid-rise buildings are usually surrounded by buildings of different types, which indicate that simply fusing the prediction maps of each scene-specific decoder network is not yet effective in extracting buildings

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

TABLE V
QUANTITATIVE COMPARISON OF THE SINGLE-MTPA-NET AND MTPA-NET

Scene	Model	IoU	Accuracy	Precision	Recall	F1-score
Urban Villages	Single-MTPA	0.874	**0.9182**	**0.9351**	0.9306	0.9327
	MTPA	**0.8756**	0.9172	0.9119	**0.9567**	**0.9337**
Industrial Buildings	Single-MTPA	**0.674**	**0.8551**	**0.7914**	0.8208	**0.804**
	MTPA	0.6727	0.8415	0.7272	**0.9027**	0.8025
Mid-rise Buildings	Single-MTPA	**0.7795**	**0.9039**	**0.8828**	0.8704	**0.8743**
	MTPA	0.7623	0.893	0.8583	**0.8749**	0.8637
High-rise Buildings	Single-MTPA	0.6817	0.9369	**0.7927**	0.8297	0.8107
	MTPA	**0.6969**	**0.9377**	0.7709	**0.879**	**0.8214**
Rural Villages	Single-MTPA	0.6632	0.8886	**0.8138**	0.782	0.7975
	MTPA	**0.7133**	**0.9039**	0.8136	**0.8532**	**0.8327**
Low-Density Buildings	Single-MTPA	0.7549	0.945	0.8658	0.854	0.8596
	MTPA	**0.7931**	**0.9528**	**0.8748**	**0.8938**	**0.8839**
High Density Buildings	Single-MTPA	0.6691	0.8904	**0.8183**	0.7857	0.8016
	MTPA	**0.7212**	**0.9055**	0.8094	**0.8689**	**0.8379**
Overall	Single-MTPA	0.7411	0.9071	**0.8562**	0.8457	0.8505
	MTPA	**0.7625**	**0.9118**	0.8329	**0.8992**	**0.8644**

from highly mixed scenes. Nevertheless, it is worth noting that the main difference relies on the number of false negatives. When we adopted the scene prior to building extraction, we can use different model parameters for buildings in different scenes. As a result, the recall of the MTPA-Net increases from 84.57% to 89.92%. This fact further demonstrates that combining scene prior with building extraction can help to reduce the number of false alarms and increase the accuracy of building extraction. It can be inferred that given a real-world data set, the Single-MTPA-Net treats each building equally, ignorant of the contextual information provided by different scenes, resulting in a high accuracy in urban scenes and low accuracy in others. Conversely, if we add scene prior to the process of building extraction, the algorithm is more robust to buildings in different scenes and can achieve better building extraction results in a real-world data set.

E. Ablation Experiments

To further explore the contributions of different modules (e.g., atrous convolution, attention mechanisms, and auxiliary classifiers) and training strategies (e.g., the Ranger optimizer and Lovasz-Softmax loss) in improving the performance of the MTPA-Net, we conducted ablation experiments on the data set. We first constructed the baseline model on the basis of the MTPA-Net. In the baseline, the auxiliary classifiers were removed, each attention module was replaced with a 1×1 convolution layer and the dilation rate of each convolutional layer was set to 1. Moreover, while training the baseline, SGD and cross-entropy loss were chosen as the optimizer algorithm and loss function, respectively.

On the basis of the baseline, optimizing the parameters of the network with the Ranger optimizer is represented by (R). The auxiliary classifiers that optimize the learning process by ensuring the effectiveness of multiscale features are named (A), and the attention modules, including CBAM and DA module, are called (M). The atrous convolution in DownRes-a and UpRes-a modules is represented by (D). Moreover, the SOTA Lovasz-Softmax loss function is represented by (L). Performance was evaluated on IoU, Precision, Recall, and F1 score.

The results of the ablation study are shown in Table VI, and the bolded values represent the highest values of the evaluation metrics among the methods. On the basis of the baseline, the quantitative evaluation of the contributions of different modules and strategies is shown in Table VI The results show that convolutional layers with different dilation rates in DownRes-a and UpRes-a modules contribute 0.4% to the increment in overall accuracy, which indicates that atrous convolution can slightly improve the performance of building extraction. The attention-based mechanisms, including CBAM and DA module, can obviously improve the performance in building extraction, improving IoU by 1%. Using auxiliary classifiers achieved considerable improvement in reducing false prediction, which contributed 1% to the improvement of IoU. Moreover, the Ranger optimizer and Lovasz-Softmax loss contribute 0.8% and 1% to the increment in IoU, respectively, which indicates that the model can achieve better building extraction performance by introducing SOTA training strategies.

F. Comparison Analysis

1) Scene Classification: To demonstrate the advantage of the proposed method, we also compared the performance in scene classification among MTPA-Net, SE-Net [62], and ResNet [63], using overall accuracy, precision, recall, and F1 score. ResNet and SE-Net are two SOTA networks for image classification. ResNet was proposed by He *et al.* and won the first prize in the ImageNet Challenge in 2015. The residual blocks of ResNet enable the gradient to flow through the entire network and are effective in avoiding the vanishing gradient and exploding the gradient phenomenon. To realize multilabel classification, we replaced the softmax operation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
INFLUENCE OF DIFFERENT MODULES AND STRATEGIES. D, ATROUS CONVOLUTION; M, ATTENTION MECHANISMS; A, AUXILIARY CLASSIFIER; L, LOVASZ-SOFTMAX; R, RANGER OPTIMIZER

Method	IoU(%)	Accuracy(%)	Precision(%)	Recall(%)	F1-score(%)	Rise IoU(%)
Baseline	73.2	90.1	83.2	85.9	84.5	×
Baseline+D	73.4	90.5	**86.1**	83.2	84.6	0.2
Baseline+DM	74.4	90.7	84.7	85.9	85.2	1.2
Baseline+DMA	75.4	90.5	83	86	85.9	2.2
Baseline+DML	75.4	90.5	82.8	89.4	85.9	2.2
Baseline+MR	75.2	90.4	82.8	89.1	85.7	2.0
MTPA-Net	**76.3**	**91.2**	83.3	**89.9**	**86.4**	**3.1**

TABLE VII
QUANTITATIVE COMPARISON OF SCENE CLASSIFICATION WITH THE SOTA METHODS

Method	Accuracy	Precision	Recall	F1-score
MTPA-Net	**0.977**	0.914	**0.943**	**0.928**
SE-Net	0.972	**0.936**	0.873	0.903
ResNet-101	0.954	0.882	0.8	0.839

with a sigmoid operation to generate the multilabel scene classification results. More details of ResNet can be found in [63]. SE-Net was proposed by Hu *et al.* and won the first prize in the ImageNet Challenge in 2017. Based on residual learning, SE-Net introduced a squeeze-and-excitation module to capture the channel-wise attention and enhance the informative channels based on the attention vector. In addition, we adopted the sigmoid function as an activation function to generate the multilabel scene classification results. More details of the SE-Net can be found in [62].

The results are shown in Table VII, and the bolded values represent the highest values of the evaluation metrics among the three methods. As shown in Table VII, the MTPA-Net achieves 0.5% and 2.3% better performance in overall accuracy than SE-Net and ResNet, respectively. This indicates that the high-level features extracted from the encoder network are useful for multilabel scene classification. Moreover, there is an approximately 2.5% increment in F1 score compared with SE-Net, which confirms that feature maps extracted from the encoder network can be shared between the classification and segmentation tasks, because the frozen parameters in the encoder network are initialized by the Single-MTPA-Net, which is trained only for building extraction.

2) Building Extraction: Seven SOTA building extraction methods, including MBI [13], SegNet [49], DeeplabV3+ [47], PSPNet [48], Res-U-Net [50], MA-FCN [52], and BRR-Net [16] were used for comparisons. These seven methods are widely adopted and have already shown their effectiveness in semantic segmentation and/or building extraction from VHR remote sensing imagery. Here, we provide a brief summary of the characteristics of each method.

a) MBI: The MBI is a handcrafted feature-based building extraction method that was proposed by Huang and Zhang [13]. The MBI aims to construct a relationship between the different features of buildings and the properties of morphological operators. After constructing the MBI image, 0.12 was adopted as the threshold to generate the binary building map in our experiment.

b) SegNet: SegNet is a fundamental FCN architecture for semantic pixel-wise segmentation introduced by Badrinarayanan *et al.* [44]. It has an encoder–decoder architecture and several skip connections. Moreover, in the decoder part, upsampling is performed using the pooling indices in the encoding component. Overall, SegNet is a widely used baseline in the performance evaluation of semantic segmentation owing to its advantages of high efficiency.

c) DeeplabV3+: Chen *et al.* [45] designed a novel encoder–decoder architecture named DeeplabV3+, which is a new masterpiece of Deeplab FCNs. The DeeplabV3+ model combines the advantages of encoding multiscale context information and capturing sharper boundaries by an improved Xception-41 encoder, atrous spatial pyramid pooling, and decoder module. Deeplab V3+ achieved SOTA performance in the PASCAL VOC 2012 benchmark.

d) PSPNet: Zhao *et al.* [48] proposed PSPNet for semantic segmentation and scene parsing. In this model, a novel pyramid pooling module is introduced to embed different subregion context information. Moreover, an effective optimization strategy is proposed based on deeply supervised loss.

e) Res-U-Net: Xu *et al.* [50] designed a U-Net-based architecture for building extraction. Input images of Res-U-Net include red, green, blue, and near-infrared bands of imagery, as well as handcrafted features such as NDVI and nDSM. A deep residual network is designed in the decoder part and a guided filter is used to generate the final prediction.

f) MA-FCN: Wei *et al.* [52] proposed a scale robust MA-FCN for automatic building footprint extraction. MA-FCN is capable of aggregating multiscale feature pyramids from different convolution layers. It generates building prediction maps on different scales and further aggregates the prediction maps with a convolution layer to generate the final prediction map.

g) BRRNet: Shao *et al.* [16] designed BRRNet for automatic building extraction in 2020. The network consists of an encoder–decoder structure-based prediction module and a novel residual refinement module. The prediction module generates the building extraction results of the input image.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

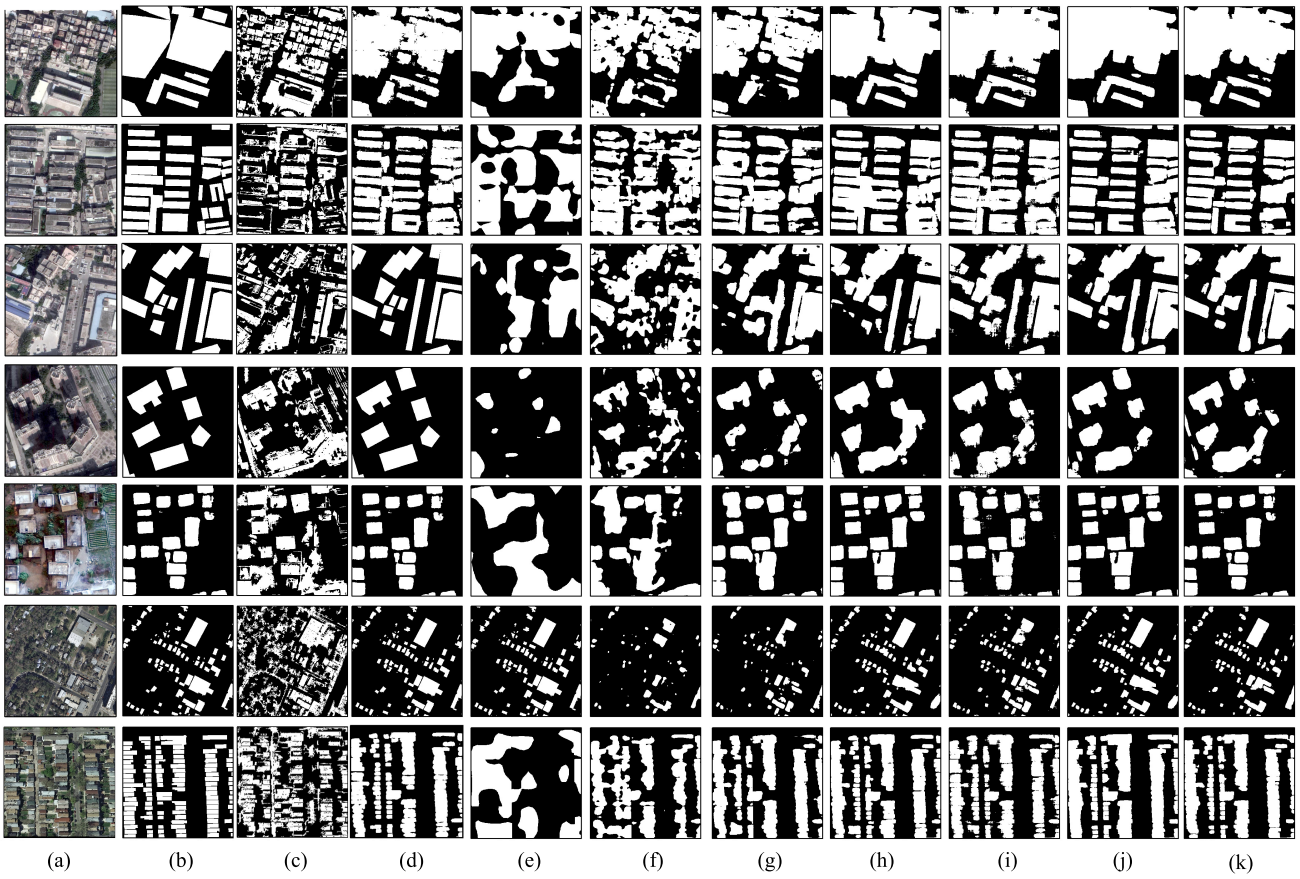


Fig. 12. Examples of the building extraction results by our proposed MTPA-Net, Single-MTPA-Net, and various SOTA models on the test data set. The first and second columns are aerial image and ground truth, respectively. Results of columns (c)–(k) are the building extraction results of MBI, SegNet, PSPNet, DeeplabV3+, Res-U-Net, MA-FCN, GRRNet, and our proposed MTPA-Net, respectively. (a) Original image. (b) Ground truth. (c) MBI. (d) SegNet. (e) PSPNet. (f) Deeplab v3+. (g) Res-U-Net. (h) MA-FCN. (i) GRRNet. (j) Single-MTPA-Net. (k) MTPA-Net.

Then, the residual refinement module further improves the results using several residual atrous convolution layers.

Fig. 12 displays part of the test images and their outcomes using different deep models across different scenes. As can be seen in Fig. 12, the MTPA-Net obtains better classification results compared with other methods visually. The improvements lie in recognizing and localizing multiscale buildings more accurately while preserving high accuracy among different scenes, which benefits from the task-specific features learned from each encoder network. In comparison, Deeplab v3+ and SegNet performed well in low-density buildings and high-density buildings than in the urban areas of Shenzhen, where building density is rather high and it is difficult to predict accurate building boundaries. Among the SOTA methods, MBI performed the worst visually. There were many false-negative predictions for roads that were similar to buildings in material and color. Moreover, the building extraction results of MBI are not robust enough to cope with buildings in different complex scenes. Among the deep learning-based methods, PSPNet also did not perform well by generating predictions with a smooth boundary, which indicates that the pyramid pooling module is not suitable for building extraction from VHR remote sensing imagery. This is mainly because the PSPNet aggregates different contextual information by a pyramid pooling module before making the final prediction, which usually produces smoother prediction results and is not suitable for building extraction. MA-FCN performed the best among the seven comparison results, with rather high accuracy in extracting urban villages and low-density buildings. However, there are some inconsistencies in the prediction results of MA-FCN, which can be explained by the fact that the inception field of MA-FCN is not large enough to cover the entire building, resulting in large holes in large-scale buildings. Moreover, there are some FNs in the building extraction results of SegNet and MTPA-Net in high-rise buildings, whereas there are many FPs in the building extraction results of Res-U-Net and PSPNet. This is because high-rise buildings have complex context information such as shadows and are tilted in images. GRR-Net performed well in extracting buildings from urban scenes but not sufficiently well to extract buildings in rural and urban fringe areas.

The quantitative results of the MTPA-Net and other comparison methods are summarized in Table VIII, which indicates the outstanding performance of the MTPA-Net over other SOTA FCNs. The bolded values in Table VII represent the highest values of the evaluation metrics among the methods. The MTPA-Net generated the best building extraction results with an IoU of 0.7625 and an overall accuracy of 0.9118 among all comparison methods. Compared with Res-U-Net, the MTPA-Net achieved approximately 1%, 1%,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
QUANTITATIVE COMPARISON OF BUILDING EXTRACTION WITH THE SOTA METHODS

Method	IoU	Accuracy	Precision	Recall	F1-score
MTPA-Net	**0.7625**	**0.9118**	0.8329	**0.8992**	**0.8644**
Single-MTPA-Net	0.7411	0.9071	**0.8562**	0.8457	0.8505
Res-U-Net	0.7349	0.9017	0.8264	0.868	0.8464
DeeplabV3+	0.7269	0.8981	0.8204	0.8638	0.8411
PSP Net	0.6983	0.8828	0.7852	0.8618	0.8214
SegNet	0.6322	0.8741	0.8315	0.7214	0.7683
MA-FCN	0.7389	0.9042	0.8293	0.8714	0.8498
BRRNet	0.7289	0.9002	0.8249	0.8622	0.8432
MBI	0.3281	0.7582	0.4797	0.5094	0.4941

TABLE IX
COMPLEXITY COMPARISON WITH THE SOTA METHODS

Method	Parameters (M)	FLOPs (G)	Model Size (MB)
Single-MTPA-Net	**13.94**	53.8	**54.66**
MTPA-Net	47.9	78.7	188.04
MA-FCN	24.08	196.4	94.1
BRRNet	17.33	244.5	67.78
Res-U-Net	24.9	**41.68**	97.27
Deeplab V3+	53.6	82.7	210.42
PSPNet	74.7	159.33	292.26
SegNet	29.44	160.41	115.11

3%, and 2% relative increase in the value of IoU, overall accuracy, precision, recall, and F1 score, respectively, which indicated that our proposed method performed well in suppressing false negatives and false positives. The improvements in the results of the MTPA-Net confirm that it can achieve better performance by collaborating building extraction with the contextual information provided by the scene prior. It should be noted that the Single-MTPA-Net performed the best among other SOTA FCNs except for the MTPA-Net, which demonstrated the effectiveness of enlarging model's reception field and emphasizing the meaningful information. Although there are as many as four parallel branches in the DownRes-a block and UpRes-a block, atrous convolution is simple but effective in enlarging the reception field while maintaining low memory usage.

G. Complexity of the MTPA-Net

To validate the tradeoff between the performance and complexity of our proposed MTPA-Net, we compared the parameters, floating point of operations (FLOPs), and model size of the eight deep learning-based methods on the data set. As shown in Table IX, PSPNet is the most complicated model with the most parameters and the largest model size. However, the performance of PSPNet was rather poor. The proposed Single-MTPA-Net has the least parameters and smallest model size, which indicates that it is a lightweight and effective model for building extraction. Compared with the Single-MTPA-Net, the increase of parameters in the MTPA-Net mainly lies in the scene classification network and multiple scene-specific decoder networks. Although SegNet, BRRNet, and MA-FCN have fewer parameters than our proposed MTPA-Net, their FLOPs are higher. This is mainly because in the MTPA-Net, not all decoder networks propagate forward, and the number of decoders involved in propagation calculation is equal to the number of scene categories included in the input image. BRRNet has the largest FLOPs, 244.5G. This is because BRRNet maintains higher resolution feature maps, which may lead to a high number of FLOPs.

V. CONCLUSION

There is a large in-class variance in buildings' color, context, and background among different scenes in the VHR remote sensing imagery, which indicates that there exist semantic gaps among different types of buildings. If we perform different building extraction strategies according to different scene categories, we may be able to achieve improved extraction results. With this motivation, we propose a novel MTPA-Net for building footprint extraction. The proposed MTPA-Net framework is an attempt to overcome the semantic gaps in building extraction by introducing the concept of "scene." In the MTPA-Net, we first classify the input image into several different scene categories. Afterward, the scene-specific decoder networks predict classification maps based on the scene-prior information provided by the classification network. The MTPA-Net is a multitask learning framework with attention modules, which is capable of learning both task-shared and task-specific features in an end-to-end manner. Experiments on a real-world data set show that the MTPA-Net can effectively increase the accuracy of building extraction, which demonstrates the superiority of incorporating scene context information into the process of building extraction. In our future research, we plan to improve the scene classification network by performing pixel-level scene classification, which will provide richer scene context information for accurately extracting buildings from VHR imagery.

REFERENCES

[1] S. Xu *et al.*, "Automatic building rooftop extraction from aerial images via hierarchical RGB-D priors," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 7369–7387, Dec. 2018, doi: 10.1109/TGRS.2018.2850972.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

[2] M. M. Rathore, A. Ahmad, A. Paul, and S. Rho, "Urban planning and building smart cities based on the Internet of Things using big data analytics," *Comput. Netw.*, vol. 101, pp. 63–80, Jun. 2016.

[3] F. Rottensteiner, "Automatic generation of high-quality building models from lidar data," *IEEE Comput. Graph. Appl.*, vol. 23, no. 6, pp. 42–50, Nov. 2003.

[4] P. Liu *et al.*, "Building footprint extraction from high-resolution images via spatial residual inception convolutional neural network," *Remote Sens.*, vol. 11, no. 7, p. 830, Apr. 2019.

[5] J. Tian, S. Cui, and P. Reinartz, "Building change detection based on satellite stereo imagery and digital surface models," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 406–417, Jan. 2014, doi: 10.1109/TGRS.2013.2240692.

[6] L. Wang, P. D. Groves, and M. K. Ziebart, "GNSS shadow matching: Improving urban positioning accuracy using a 3D city model with optimized visibility scoring scheme," *Navigation*, vol. 60, no. 3, pp. 195–207, Sep. 2013.

[7] Y. Liu *et al.*, "Multilevel building detection framework in remote sensing images based on convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 10, pp. 3688–3700, Oct. 2018, doi: 10.1109/JSTARS.2018.2866284.

[8] J. Huang, X. Zhang, Q. Xin, Y. Sun, and P. Zhang, "Automatic building extraction from high-resolution aerial images and LiDAR data using gated residual refinement network," *ISPRS J. Photogramm. Remote Sens.*, vol. 151, pp. 91–105, May 2019, doi: 10.1016/j.isprsjprs.2019.02.019.

[9] M. Cote and P. Saeedi, "Automatic rooftop extraction in nadir aerial imagery of suburban regions using corners and variational level set evolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 1, pp. 313–328, Jan. 2013, doi: 10.1109/TGRS.2012.2200689.

[10] Y. Zha, J. Gao, and S. Ni, "Use of normalized difference built-up index in automatically mapping urban areas from TM imagery," *Int. J. Remote Sens.*, vol. 24, no. 3, pp. 583–594, Jan. 2003, doi: 10.1080/01431160304987.

[11] T. Zhang, X. Huang, D. Wen, and J. Li, "Urban building density estimation from high-resolution imagery using multiple features and support vector regression," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 7, pp. 3265–3280, Jul. 2017, doi: 10.1109/JSTARS.2017.2669217.

[12] Z. Sun, H. Fang, M. Deng, A. Chen, P. Yue, and L. Di, "Regular shape similarity index: A novel index for accurate extraction of regular objects from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3737–3748, Jul. 2015, doi: 10.1109/TGRS.2014.2382566.

[13] X. Huang and L. Zhang, "A multidirectional and multiscale morphological index for automatic building extraction from multispectral GeoEye-1 imagery," *Photogramm. Eng. Remote Sens.*, vol. 77, no. 7, pp. 721–732, Jul. 2011.

[14] M. Awrangjeb, C. Zhang, and C. S. Fraser, "Improved building detection using texture information," *ISPRS-Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 38, pp. 143–148, Apr. 2013.

[15] G. Zhou, "Building extraction from LIDAR data based on shape analysis of contours," *J. Southwest Jiaotong Univ.*, vol. 44, no. 1, pp. 83–88, 2009.

[16] Z. Shao, P. Tang, Z. Wang, N. Saleem, S. Yam, and C. Sommai, "BRRNet: A fully convolutional neural network for automatic building extraction from high-resolution remote sensing images," *Remote Sens.*, vol. 12, no. 6, p. 1050, Mar. 2020.

[17] J. Niemeyer, F. Rottensteiner, and U. Soergel, "Contextual classification of lidar data and building object detection in urban areas," *ISPRS J. Photogramm. Remote Sens.*, vol. 87, pp. 152–165, Jan. 2014.

[18] L. Ma, Y. Liu, X. Zhang, Y. Ye, G. Yin, and B. A. Johnson, "Deep learning in remote sensing applications: A meta-analysis and review," *ISPRS J. Photogramm. Remote Sens.*, vol. 152, pp. 166–177, Jun. 2019.

[19] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. NIPS*, 2012, pp. 1097–1105.

[20] G. Wu *et al.*, "Automatic building segmentation of aerial imagery using multi-constraint fully convolutional networks," *Remote Sens.*, vol. 10, no. 3, p. 407, Mar. 2018, doi: 10.3390/rs10030407.

[21] L. Sun, Y. Tang, and L. Zhang, "Rural building detection in high-resolution imagery based on a two-stage CNN model," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 1998–2002, Nov. 2017, doi: 10.1109/LGRS.2017.2745900.

[22] Q. Shi *et al.*, "Domain adaption for fine-grained urban village extraction from satellite images," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 8, pp. 1430–1434, Aug. 2020, doi: 10.1109/LGRS.2019.2947473.

[23] X. Jin and C. H. Davis, "Automated building extraction from high-resolution satellite imagery in urban areas using structural, contextual, and spectral information," *EURASIP J. Adv. Signal Process.*, vol. 2005, no. 14, pp. 2196–2206, Dec. 2005.

[24] W. Luo, H. Li, G. Liu, and L. Zeng, "Semantic annotation of satellite images using author–genre–topic model," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 2, pp. 1356–1368, Feb. 2014.

[25] Y. Zhong, Q. Zhu, and L. Zhang, "Scene classification based on the multifeature fusion probabilistic topic model for high spatial resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 11, pp. 6207–6222, Nov. 2015.

[26] Y. Huang, Z. Wu, L. Wang, and T. Tan, "Feature coding in image classification: A comprehensive study," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 3, pp. 493–506, Mar. 2014.

[27] B. Zhao, Y. Zhong, L. Zhang, and B. Huang, "The Fisher kernel coding framework for high spatial resolution scene classification," *Remote Sens.*, vol. 8, no. 2, p. 157, Feb. 2016. [Online]. Available: https://www.mdpi.com/2072-4292/8/2/157

[28] Q. Zhu, Y. Zhong, L. Zhang, and D. Li, "Adaptive deep sparse semantic modeling framework for high spatial resolution image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6180–6195, Oct. 2018, doi: 10.1109/TGRS.2018.2833293.

[29] Y. Liu and C. Huang, "Scene classification via triplet networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 1, pp. 220–237, Jan. 2018.

[30] C. Tao, L. Mi, Y. Li, J. Qi, Y. Xiao, and J. Zhang, "Scene context-driven vehicle detection in high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7339–7351, Oct. 2019, doi: 10.1109/TGRS.2019.2912985.

[31] X. Huang and L. Zhang, "Morphological building/shadow index for building extraction from high-resolution imagery over urban areas," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 1, pp. 161–172, Feb. 2012.

[32] B. Sirmacek and C. Unsalan, "Urban-area and building detection using SIFT keypoints and graph theory," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 4, pp. 1156–1167, Apr. 2009, doi: 10.1109/TGRS.2008.2008440.

[33] G.-S. Xia, J. Huang, N. Xue, Q. Lu, and X. Zhu, "GeoSay: A geometric saliency for extracting buildings in remote sensing images," *Comput. Vis. Image Understand.*, vol. 186, pp. 37–47, Sep. 2019, doi: 10.1016/j.cviu.2019.06.001.

[34] S. Du, Y. Zhang, Z. Zou, S. Xu, X. He, and S. Chen, "Automatic building extraction from LiDAR data fusion of point and grid-based features," *ISPRS J. Photogramm. Remote Sens.*, vol. 130, pp. 294–307, Aug. 2017, doi: 10.1016/j.isprsjprs.2017.06.005.

[35] R. Wang, Y. Hu, H. Wu, and J. Wang, "Automatic extraction of building boundaries using aerial LiDAR data," *J. Appl. Remote Sens.*, vol. 10, no. 1, Mar. 2016, Art. no. 016022, doi: 10.1117/1.JRS.10.016022.

[36] K. Zhang, J. Yan, and S.-C. Chen, "Automatic construction of building footprints from airborne LIDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 9, pp. 2523–2533, Sep. 2006, doi: 10.1109/TGRS.2006.874137.

[37] A. Zarea and A. Mohammadzadeh, "A novel building and tree detection method from LiDAR data and aerial images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 5, pp. 1864–1875, May 2016, doi: 10.1109/JSTARS.2015.2470547.

[38] X. Meng, N. Currit, L. Wang, and X. Yang, "Detect residential buildings from lidar and aerial photographs through object-oriented land-use classification," *Photogramm. Eng. Remote Sens.*, vol. 78, no. 1, pp. 35–44, Jan. 2012.

[39] Q. Wang, S. Liu, J. Chanussot, and X. Li, "Scene classification with recurrent attention of VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1155–1167, Feb. 2019.

[40] Q. Li, L. Mou, Q. Xu, Y. Zhang, and X. X. Zhu, "R3-Net: A deep network for multioriented vehicle detection in aerial images and videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 5028–5042, Jul. 2019, doi: 10.1109/TGRS.2019.2895362.

[41] G. Zhang, S. Lu, and W. Zhang, "CAD-Net: A context-aware detection network for objects in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10015–10024, Dec. 2019, doi: 10.1109/TGRS.2019.2930982.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.

[42] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[43] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.

[44] V. Badrinarayanan, A. Kendall, and R. Cipolla, "Segnet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[45] L. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "Semantic image segmentation with deep convolutional nets and fully connected CRFs," 2014, *arXiv:1412.7062*. [Online]. Available: https://arxiv.org/abs/1412.7062

[46] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*. [Online]. Available: http://arxiv.org/abs/1706.05587

[47] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," 2018, *arXiv:1802.02611*. [Online]. Available: http://arxiv.org/abs/1802.02611

[48] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," 2016, *arXiv:1612.01105*. [Online]. Available: http://arxiv.org/abs/1612.01105

[49] R. Alshehhi, P. R. Marpu, W. L. Woon, and M. D. Mura, "Simultaneous extraction of roads and buildings in remote sensing imagery with convolutional neural networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 130, pp. 139–149, Aug. 2017.

[50] Y. Xu, L. Wu, Z. Xie, and Z. Chen, "Building extraction in very high resolution remote sensing imagery using deep learning and guided filters," *Remote Sens.*, vol. 10, no. 1, p. 144, Jan. 2018. [Online]. Available: https://www.mdpi.com/2072-4292/10/1/144

[51] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019, doi: 10.1109/TGRS.2018.2858817.

[52] S. Wei, S. Ji, and M. Lu, "Toward automatic building footprint delineation from aerial images using CNN and regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 2178–2189, Mar. 2020, doi: 10.1109/TGRS.2019.2954461.

[53] H. Liu *et al.*, "DE-Net: Deep encoding network for building extraction from high-resolution remote sensing imagery," *Remote Sens.*, vol. 11, no. 20, p. 2380, Oct. 2019. [Online]. Available: https://www.mdpi.com/2072-4292/11/20/2380

[54] J. Fu, J. Liu, H. Tian, Y. Li, Y. Bao, Z. Fang, and H. Lu, "Dual attention network for scene segmentation," 2018, *arXiv:1809.02983*. [Online]. Available: http://arxiv.org/abs/1809.02983

[55] S. Woo, J. Park, J.-Y. Lee, and I. So Kweon, "CBAM: Convolutional block attention module," 2018, *arXiv:1807.06521*. [Online]. Available: http://arxiv.org/abs/1807.06521

[56] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Can semantic labeling methods generalize to any city? The inria aerial image labeling benchmark," in *Proc. IGARSS-IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2017, pp. 3226–3229.

[57] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," 2019, *arXiv:1912.01703*. [Online]. Available: http://arxiv.org/abs/1912.01703

[58] L. Liu *et al.*, "On the variance of the adaptive learning rate and beyond," 2019, *arXiv:1908.03265*. [Online]. Available: http://arxiv.org/abs/1908.03265

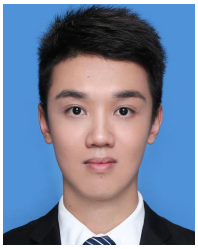
[59] M. R. Zhang, J. Lucas, G. Hinton, and J. Ba, "Lookahead optimizer: K steps forward, 1 step back," 2019, *arXiv:1907.08610*. [Online]. Available: http://arxiv.org/abs/1907.08610

[60] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on imagenet classification," 2015, *arXiv:1502.01852*. [Online]. Available: http://arxiv.org/abs/1502.01852

[61] M. Berman, A. R. Triki, and M. B. Blaschko, "The Lovász-Softmax loss: A tractable surrogate for the optimization of the intersection-over-union measure in neural networks," 2017, *arXiv:1705.08790*. [Online]. Available: https://arxiv.org/abs/1705.08790 and https://ui.adsabs.harvard.edu/abs/2017arXiv170508790B

[62] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," 2017, *arXiv:1709.01507*. [Online]. Available: http://arxiv.org/abs/1709.01507

[63] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," 2015, *arXiv:1512.03385*. [Online]. Available: http://arxiv.org/abs/1512.03385



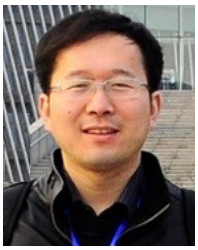
Haonan Guo received the B.S. degree from Sun Yat-sen University, Guangzhou, China, in 2020. He is pursuing the M.S. degree with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, Wuhan, China.

His research interests include deep learning, building footprint extraction, urban remote sensing, and object detection.



Qian Shi (Senior Member, IEEE) received the B.S. degree in sciences and techniques of remote sensing from Wuhan University, Wuhan, China, in 2010 and the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, in 2015.

She is an Associate Professor with the School of Geography and Planning, Sun Yat-sen University, Guangzhou, China. Her research interests include remote sensing image classification, including deep learning, active learning, and transfer learning.



Bo Du (Senior Member, IEEE) received the B.S. degree and the Ph.D. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote sensing, Wuhan University, Wuhan, China, in 2005 and 2010, respectively.

He is a Professor with the School of Computer, Wuhan University. He has authored more than 60 research articles published in the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), the IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), the IEEE JOURNAL OF SELECTED TOPICS IN EARTH OBSERVATIONS AND APPLIED REMOTE SENSING (JSTARS), and the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL). Thirteen of them are ESI hot articles or highly cited articles. His major research interests include pattern recognition, hyperspectral image processing, and signal processing.

Dr. Du received the Distinguished Paper Award from IJCAI 2018, the Best Paper Award of the IEEE ICDM DELL Workshop 2018, the Best Paper Award of the IEEE Whispers 2018, and the Champion Award of the IEEE Data Fusion Contest 2018. He received the best reviewer awards from the IEEE GRSS for his service to the JSTARS in 2011 and ACM rising star awards for his academic progress in 2015. He was the Session Chair for both the International Geoscience and Remote Sensing Symposium (IGARSS) 2018/2016 and the 4th IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing (WHISPERS). He also serves as a reviewer for 20 Science Citation Index (SCI) magazines including the IEEE TGRS, TIP, JSTARS, and GRSL. He is serving as an Associate Editor of *Pattern Recognition* and *Neurocomputing*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.



Liangpei Zhang (Fellow, IEEE) received the B.S. degree in physics from Hunan Normal University, Changsha, China, in 1982, the M.S. degree in optics from the Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China, in 1988, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 1998.

He was the Head of the Remote Sensing Division, State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University. He is also a "Chang-Jiang Scholar" Chair Professor appointed by the ministry of education of China. He was a Principal Scientist for the China state key basic research project from 2011 to 2016 appointed by the Ministry of National Science and Technology of China to lead the remote sensing program in China. He has authored more than 700 research articles and 6 books. He holds 30 patents. His research interests include hyperspectral remote sensing, high-resolution remote sensing, image processing, and artificial intelligence.

Dr. Zhang is a Fellow of the Institution of Engineering and Technology (IET) and an Executive Member (Board of Governors) of the China National Committee of International Geosphere–Biosphere Programme and the China Society of Image and Graphics. He was a recipient of the 2010 Best Paper Boeing Award and the 2013 Best Paper ERDAS Award from the American Society of Photogrammetry and Remote Sensing (ASPRS). He is the Founding Chair of the IEEE Geoscience and Remote Sensing Society (GRSS) Wuhan Chapter. He received the Best Reviewer Award from the IEEE GRSS for his service to the IEEE JOURNAL OF SELECTED TOPICS IN EARTH OBSERVATIONS AND APPLIED REMOTE SENSING (JSTARS) in 2012 and the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL) in 2014. He was the General Chair for the 4th IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing (WHISPERS) and the Guest Editor of JSTARS. His research teams won the top three prizes of the IEEE GRSS 2014 Data Fusion Contest, and his students have been selected as the winners or finalists of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS) student paper contest in recent years. He regularly serves as a Co-Chair of the series SPIE conferences on multispectral image processing and pattern recognition, conference on Asia remote sensing, and many other conferences. He has been an Editor for several conference proceedings, issues, and geoinformatics symposiums. He also serves as an Associate Editor of the *International Journal of Ambient Computing and Intelligence*, the *International Journal of Image and Graphics*, the *International Journal of Digital Multimedia Broadcasting*, the *Journal of Geo-spatial Information Science*, and the *Journal of Remote Sensing*, and the Guest Editor of the *Journal of Applied Remote Sensing* and the *Journal of Sensors*. He is currently serving as an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

Dongzhi Wang, photograph and biography not available at the time of publication.

Huaxiang Ding, photograph and biography not available at the time of publication.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:28:46 UTC from IEEE Xplore. Restrictions apply.